

Number › Fractions › Simplifying Fractions (Foundation)

Topic 1 of 313 • id: fractions_simplify_F

Q1. Simplify $\frac{3}{18}$.

Number › Fractions › Equivalent Fractions (Foundation)

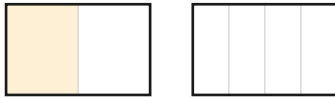
Topic 2 of 313 • id: fractions_equivalent_F

Q1. $\frac{7}{10} = \frac{?}{20}$. Find the missing number.

Number › Fractions › Equivalent Fractions (Diagrams) (Foundation)

Topic 3 of 313 • id: fractions_equivalent_diagram_F

Q1. How many parts of Shape B must be shaded to show a fraction equivalent to Shape A?



Number › Fractions › Ordering Fractions (Foundation)

Topic 4 of 313 • id: fractions_ordering_F

Q1. Write these fractions in descending order: $\frac{3}{4}$, $\frac{1}{4}$, $\frac{5}{9}$, $\frac{1}{2}$

Number › Fractions › Adding & Subtracting Fractions (Foundation)

Topic 5 of 313 • id: fractions_add_subtract_F

Q1. Work out $\frac{3}{7} - \frac{1}{4}$. Give your answer as a fraction in its simplest form.

Number › Fractions › Multiplying Fractions (Foundation)

Topic 6 of 313 • id: fractions_multiply_F

Q1. Calculate $\frac{2}{8} \times \frac{2}{7}$. Give your answer as a fraction in its simplest form.

Number › Fractions › Dividing Fractions (Higher)

Topic 7 of 313 • id: fractions_divide_H

Q1. Work out $\frac{5}{2} \div \frac{8}{10}$. Give your answer as a fraction in its simplest form.

Number › Fractions › Dividing Fractions (Foundation)

Topic 8 of 313 • id: fractions_divide_F

Q1. Evaluate $\frac{5}{2} \div \frac{3}{8}$. Give your answer as a fraction in its simplest form.

Number › Fractions › Mixed Number Arithmetic (Higher)

Topic 9 of 313 • id: fractions_mixed_number_arithmetic_H

Q1. Calculate $5\frac{3}{6} + 1\frac{1}{3}$. Give your answer as a mixed number.

Number › Fractions › Improper Fractions and Mixed Numbers (Foundation)

Topic 10 of 313 • id: fractions_improper_mixed_F

Q1. Convert $\frac{9}{5}$ to a mixed number.

Number › Fractions › Fractions of an Amount (Foundation)

Topic 11 of 313 • id: fractions_of_amount_F

Q1. Calculate $\frac{6}{9}$ of 108.

Number › Decimals › Rounding to Decimal Places (Foundation)

Topic 12 of 313 • id: decimals_round_dp_F

Q1. Round 46.341 to 1 decimal place.

Number › Decimals › Rounding to Significant Figures (Foundation)

Topic 13 of 313 • id: decimals_round_sf_F

Q1. Round 546.327 to 2 significant figures.

Number › Decimals › Ordering Decimals, Fractions & Percentages (Foundation)

Topic 14 of 313 • id: decimals_ordering_F

Q1. Write these values in descending order: 59%, 0.56, 0.46, $\frac{49}{100}$

Number › Decimals › Adding and Subtracting Decimals (Foundation)

Topic 15 of 313 • id: decimals_add_subtract_F

Q1. Evaluate $28.72 + 92.5$

Number › Decimals › Multiplying Decimals (Foundation)

Topic 16 of 313 • id: decimals_multiply_F

Q1. Work out 15.18×9.17

Number › Decimals › Dividing Decimals (Foundation)

Topic 17 of 313 • id: decimals_divide_F

Q1. Work out $0.56 \div 0.08$

Number › Decimals › Recurring Decimals to Fractions (One Digit) (Foundation)

Topic 18 of 313 • id: recurring_decimal_single_digit_F

Q1. Write $0.\dot{6}$ as a fraction in its simplest form.

Number › Decimals › Recurring Decimals to Fractions (Two Digits) (Higher)

Topic 19 of 313 • id: recurring_decimal_two_digit_H

Q1. Write $0.\dot{1}\dot{8}$ as a fraction in its simplest form.

Number › Decimals › Recurring Decimals to Fractions (Higher)

Topic 20 of 313 • id: decimals_recurring_to_fraction_H

Q1. Write $0.\dot{8}5714\dot{2}$ as a fraction in its simplest form.

Number › Order of Operations (BIDMAS) › BIDMAS (2 or 3 operations) (Foundation)

Topic 21 of 313 • id: bidmas_two_three_F

Q1. Work out $8 \times 8 - 60$.

Number › Order of Operations (BIDMAS) › BIDMAS (4+ operations) (Foundation)

Topic 22 of 313 • id: bidmas_F

Q1. Work out $2^2 + 3 \times 6 - 7 \div 7$.

Number › Standard Form › Converting to Standard Form (Foundation)

Topic 23 of 313 • id: standard_form_to_F

Q1. Write 50000000 in standard form.

Number › Standard Form › Converting Small Numbers to Standard Form (Foundation)

Topic 24 of 313 • id: standard_form_to_small_F

Q1. Write 0.00000005 in standard form.

Number › Standard Form › Converting from Standard Form (Large Numbers) (Foundation)

Topic 25 of 313 • id: standard_form_from_large_F

Q1. Write 5×10^4 as an ordinary number.

Number › Standard Form › Converting from Standard Form (Small Numbers) (Foundation)

Topic 26 of 313 • id: standard_form_from_small_F

Q1. Write 3.81×10^{-3} as an ordinary number.

Number › Standard Form › Multiplying & Dividing in Standard Form (Higher)

Topic 27 of 313 • id: standard_form_multiply_divide_H

Q1. Work out $(5 \times 10^{-4}) \div (4 \times 10^{-3})$. Give your answer in standard form.

Number › Standard Form › Multiplying & Dividing in Standard Form (Foundation)

Topic 28 of 313 • id: standard_form_multiply_divide_F

Q1. Work out $(2 \times 10^1) \times (2 \times 10^2)$. Give your answer in standard form.

Number › Standard Form › Adding & Subtracting in Standard Form (Higher)

Topic 29 of 313 • id: standard_form_add_subtract_H

Q1. Work out $(9 \times 10^6) + (3 \times 10^5)$. Give your answer in standard form.

Number › Standard Form › Standard Form with a Calculator (Foundation)

Topic 30 of 313 • id: standard_form_calculator_F

Q1. Use a calculator to work out $(9.3 \times 10^7) \div (5.9 \times 10^3)$. Give your answer in standard form, rounded to 3 significant figures if necessary.

Number › Estimation & Bounds › Estimating Calculations (Foundation)

Topic 31 of 313 • id: estimation_rounding_F

Q1. Work out an estimate for $(65.3 \times 5.4) \div 8.5$ by rounding each number to 1 significant figure.

Number › Estimation & Bounds › Error Intervals (Foundation)

Topic 32 of 313 • id: error_interval_F

Q1. A number, x , is 152 rounded to the nearest whole number. Write down the error interval for x .

Number › Estimation & Bounds › Error Intervals (significant figures) (Higher)

Topic 33 of 313 • id: error_interval_H

Q1. A number, x , is 2.4 rounded to 2 significant figures. Write down the error interval for x .

Number › Estimation & Bounds › Bounds of Calculations (Higher)

Topic 34 of 313 • id: bounds_calculation_H

Q1. A rectangle has length 9 cm and width 23 cm, both measured to the nearest cm. Calculate the lower bound for the area of the rectangle.

Number › Negative Numbers › Adding and Subtracting Negative Numbers (Foundation)

Topic 35 of 313 • id: negative_add_subtract_F

Q1. Work out $17 + (-6)$.

Number › Negative Numbers › Multiplying and Dividing Negative Numbers (Foundation)

Topic 36 of 313 • id: negative_multiply_divide_F

Q1. Work out -10×11 .

Number › Negative Numbers › Ordering Negative Numbers (Foundation)

Topic 37 of 313 • id: negative_ordering_F

Q1. Write these numbers in ascending order: -3, -8, -9, 0, 9

Number › Multiplying & Dividing by Powers of 10 › Multiplying & Dividing by Powers of 10 (Foundation)

Topic 38 of 313 • id: number_powers_of_ten_F

Q1. Evaluate $70.17 \div 100$.

Number › Factors, Multiples & Primes › Prime Numbers (Foundation)

Topic 39 of 313 • id: prime_numbers_F

From this list, write down all the prime numbers:

Q1. 33, 62, 54, 26, 14, 86

Number › Factors, Multiples & Primes › Multiples (Foundation)

Topic 40 of 313 • id: multiples_F

Q1. Is 68 a multiple of 8?

Number › Factors, Multiples & Primes › Factors (Foundation)

Topic 41 of 313 • id: factors_F

Q1. How many factors does 14 have?

Number › Factors, Multiples & Primes › Prime Factor Decomposition (Foundation)

Topic 42 of 313 • id: prime_factors_F

Q1. Write 184 as a product of its prime factors.

Number › Factors, Multiples & Primes › Prime Factor Decomposition (Index Form) (Higher)

Topic 43 of 313 • id: prime_factors_H

Q1. Write 200 as a product of its prime factors, giving your answer in index form.

Number › Factors, Multiples & Primes › LCM by Listing (Foundation)

Topic 44 of 313 • id: lcm_by_listing_F

Find the lowest common multiple (LCM) of

Q1. 2 and 8.

Number › Factors, Multiples & Primes › HCF by Listing (Foundation)

Topic 45 of 313 • id: hcf_by_listing_F

Find the highest common factor (HCF) of

Q1. 52 and 27.

Number › Factors, Multiples & Primes › HCF & LCM using Prime Factors (Higher)

Topic 46 of 313 • id: hcf_lcm_by_prime_factors_H

Using prime factorisation, find the

Q1. HCF of 134 and 55.

Number › Powers, Roots & Indices › Powers & Indices (Foundation)

Topic 47 of 313 • id: powers_F

Q1. Simplify $(4^3)^3$, giving your answer as a single power of 4.

Number › Powers, Roots & Indices › Negative Indices (Foundation)

Topic 48 of 313 • id: negative_indices_F

Q1. Work out 9^{-3} . Give your answer as a fraction.

Number › Powers, Roots & Indices › Negative & Fractional Indices (Higher)

Topic 49 of 313 • id: powers_H

Q1. Work out 6^{-2} . Give your answer as a fraction.

Number › Powers, Roots & Indices › Simplifying Indices (Challenging) (Higher)

Topic 50 of 313 • id: simplifying_indices_challenging_H

Q1. Simplify $(3^1)^2 \div 3^3$, giving your answer as a fraction where necessary.

Number › Powers, Roots & Indices › Solving Index Equations (Higher)

Topic 51 of 313 • id: indices_common_base_equations_H

Q1. Solve $9^x = 729$.

Number › Powers, Roots & Indices › Square & Cube Roots (Foundation)

Topic 52 of 313 • id: roots_F

Q1. Calculate the cube root of 729.

Number › Powers, Roots & Indices › Simplifying Surds (Higher)

Topic 53 of 313 • id: roots_H

Q1. Simplify $\sqrt{125}$, giving your answer in the form $a\sqrt{b}$.

Number › Powers, Roots & Indices › Multiplying and Dividing Surds (Higher)

Topic 54 of 313 • id: surds_multiply_divide_H

Q1. Evaluate $\sqrt{20} \div \sqrt{5}$, giving your answer as an integer.

Number › Powers, Roots & Indices › Surds in Algebraic Expressions (Higher)

Topic 55 of 313 • id: algebraic_surds_H

Q1. Expand and simplify $(7 - \sqrt{2})(2 + \sqrt{2})$, giving your answer in the form $p + q\sqrt{2}$.

Number › Powers, Roots & Indices › Rationalising the Denominator (Higher)

Topic 56 of 313 • id: rationalise_denominator_H

Q1. Rationalise the denominator of $\frac{18}{\sqrt{3}}$, giving your answer in its simplest form.

Algebra › Expressions, Formulae, Equations & Identities

› Expressions, Formulae, Equations and Identities (Foundation)

Topic 57 of 313 • id: classify_expressions_F

Is the following an expression, equation, formula, or identity?

Q1. $-7x + 7 = -13$

Algebra › Solving Linear Equations › One-Step Equations (Foundation)

Topic 58 of 313 • id: linear_one_step_F

Q1. Solve: $5x = 35$

Algebra › Solving Linear Equations › Two-Step Equations (Foundation)

Topic 59 of 313 • id: linear_two_step_F

Q1. Solve: $3x + 15 = 45$

Algebra › Solving Linear Equations › Multi-Step Equations (Foundation)

Topic 60 of 313 • id: linear_multi_step_F

Q1. Solve: $3x + 10 + 6x - 9 = 91$

Algebra › Solving Linear Equations › Unknowns on Both Sides (Foundation)

Topic 61 of 313 • id: linear_both_sides_F

Q1. Solve: $2x + 9 = 5x - 21$

Algebra › Solving Linear Equations › Unknowns on Both Sides (Higher)

Topic 62 of 313 • id: linear_both_sides_H

Q1. Solve: $7x - 9 = x - 21$

Algebra › Solving Linear Equations › Equations with Brackets (Foundation)

Topic 63 of 313 • id: linear_brackets_F

Q1. Solve: $4(3x + 4) = 100$

Algebra › Solving Linear Equations › Equations with Brackets (Higher)

Topic 64 of 313 • id: linear_brackets_H

Q1. Solve: $3(4x + 6) = 42$

Algebra › Solving Linear Equations › Equations Involving Fractions (Foundation)

Topic 65 of 313 • id: fractional_equations_F

Solve the following equation.

Q1. $\frac{x}{2} = 9$

Algebra › Solving Linear Equations › Equations Involving Fractions (Higher)

Topic 66 of 313 • id: fractional_equations_H

Solve the following equation.

Q1. $\frac{x}{2} + \frac{x}{6} = 5$

Algebra › Solving Linear Equations › Cross Multiplication (Foundation)

Topic 67 of 313 • id: cross_multiplication_F

Solve the following equation.

Q1. $\frac{7}{x} = \frac{2}{4}$

Algebra › Solving Linear Equations › Cross Multiplication (Higher)

Topic 68 of 313 • id: cross_multiplication_H

Solve the following equation.

Q1. $\frac{x-1}{6} = \frac{2x-1}{4}$

Algebra › Forming and Solving Equations › Forming and Solving Equations (Foundation)

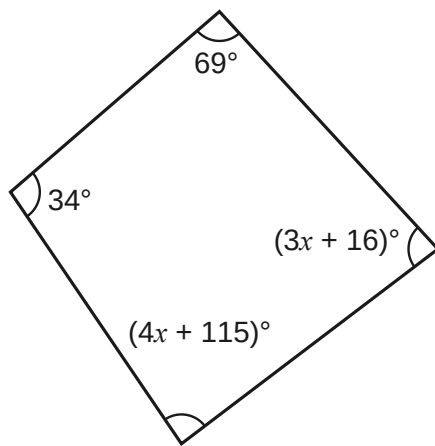
Topic 69 of 313 • id: forming_equations_F

Q1. I think of a number, multiply it by 3 and add 20. The result is 62. Form an equation and solve it to find the number.

Algebra › Forming and Solving Equations › Forming and Solving Equations (Higher)

Topic 70 of 313 • id: forming_equations_H

Q1. A quadrilateral has angles 34° , 69° , $(3x + 16)^\circ$, and $(4x + 115)^\circ$. Form an equation and solve it to find x .



Algebra › Changing the Subject of a Formula › Changing the Subject of a Formula (Foundation)

Topic 71 of 313 • id: change_subject_F

Q1. Make x the subject of the formula $y = 2x + 7$.

Algebra › Changing the Subject of a Formula › Changing the Subject of a Formula (Higher)

Topic 72 of 313 • id: change_subject_H

Q1. Make x the subject of the formula $y = 6x + 8x + 14$.

Algebra › Changing the Subject of a Formula › Changing the Subject by Factorising (Higher)

Topic 73 of 313 • id: change_subject_factorise_H

Make x the subject of the formula

Q1. $wx + 12x = 14$.

Algebra › Substitution into Formulae › Substituting into Formulae (Foundation)

Topic 74 of 313 • id: substitution_F

Q1. $A = LW$. Find the value of A when $L = 15$ and $w = 4$.

Algebra › Substitution into Formulae › Substituting into Formulae (Higher)

Topic 75 of 313 • id: substitution_H

Q1. $a = \frac{v-u}{t}$. Find the value of a when $u = 47$, $v = 54$ and $t = 7$.

Algebra › Substitution into Formulae › Rearranging and Substituting into Formulae (Foundation)

Topic 76 of 313 • id: substitution_rearrange_F

Q1. $v = u + at$. Make a the subject of the formula, then find the value of a when $v = 55$, $u = 10$ and $t = 5$.

Algebra › Substitution into Formulae › Rearranging and Substituting into Formulae (Higher)

Topic 77 of 313 • id: substitution_rearrange_H

Q1. $a = \frac{v-u}{t}$. Make u the subject of the formula, then find the value of u when $v = 41$, $a = -1$ and $t = 8$.

Algebra › Expanding Brackets › Single Bracket (Foundation)

Topic 78 of 313 • id: expand_single_bracket_F

Q1. Expand: $8(-5x - 3)$

Algebra › Expanding Brackets › Double Brackets (Foundation)

Topic 79 of 313 • id: expand_double_brackets_F

Expand and simplify:

Q1. $(4x + 7)(2x + 5)$

Algebra › Expanding Brackets › Double Brackets (No Coefficient) (Foundation)

Topic 80 of 313 • id: expand_double_brackets_no_coefficient_F

Expand and simplify:

Q1. $(x + 7)(x + 9)$

Algebra › Expanding Brackets › Double Brackets (Higher)

Topic 81 of 313 • id: expand_double_brackets_H

Expand and simplify:

Q1. $(x - 3)(4x + 3)$

Algebra › Expanding Brackets › Triple Brackets (Higher)

Topic 82 of 313 • id: expand_triple_brackets_H

Expand and simplify:

Q1. $(3x - 2)(4x - 3)(4x + 4)$

Algebra › Expanding Brackets › Triple Brackets (no coefficient) (Higher)

Topic 83 of 313 • id: expand_triple_brackets_no_coefficient_H

Expand and simplify:

Q1. $(x + 1)(x - 3)(x - 1)$

Algebra › Factorising › Common Factor (Foundation)

Topic 84 of 313 • id: factorise_common_factor_F

Q1. Factorise: $-12x - 18$

Algebra › Factorising › Quadratics (Foundation)

Topic 85 of 313 • id: factorise_quadratics_F

Q1. Factorise: $x^2 + 4x + 3$

Algebra › Factorising › Quadratics (Higher)

Topic 86 of 313 • id: factorise_quadratics_H

Q1. Factorise: $x^2 + 10x + 16$

Algebra › Solving Quadratic Equations › Solving by Factorising (Foundation)

Topic 87 of 313 • id: solve_quadratic_factorising_F

Q1. Solve $x^2 + 10x + 16 = 0$ by factorising.

Algebra › Solving Quadratic Equations › Solving by Factorising (Higher)

Topic 88 of 313 • id: solve_quadratic_factorising_H

Q1. Solve $x^2 - 9x + 20 = 0$ by factorising.

Algebra › Solving Quadratic Equations › Solving by Completing the Square (Higher)

Topic 89 of 313 • id: solve_quadratic_completing_square_H

Q1. Solve $x^2 + 18x + 70 = 0$ by completing the square, giving your answers in exact surd form.

Algebra › Solving Quadratic Equations › The Quadratic Formula (Higher)

Topic 90 of 313 • id: quadratic_formula_H

Q1. Solve $3x^2 + 13x + 13 = 0$, giving your answer in the form $x = (-b \pm k\sqrt{m})/2a$, fully simplified.

Algebra › Algebraic Indices › Laws of Indices with Algebra (Foundation)

Topic 91 of 313 • id: algebraic_indices_F

Q1. Simplify $4x^6 \times 6x^6$

Algebra › Algebraic Indices › Laws of Indices with Algebra (Negative and Fractional) (Higher)

Topic 92 of 313 • id: algebraic_indices_H

Q1. Simplify $(4x^5)^3$

Algebra › Completing the Square › Completing the Square (Higher)

Topic 93 of 313 • id: completing_the_square_H

Q1. Write $x^2 + 16x - 1$ in the form $(x + p)^2 + q$.

Algebra › Turning Point of a Graph › Turning Point of a Graph (Higher)

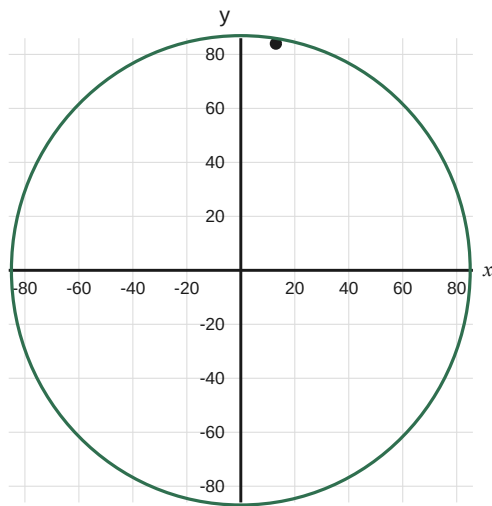
Topic 94 of 313 • id: turning_point_of_graph_H

Q1. Find the coordinates of the turning point of the curve $y = x^2 - 8x + 8$.

Algebra › Equation of a Circle › Equation of a Circle (Higher)

Topic 95 of 313 • id: circle_equation_H

Q1. The point $(13, 84)$ lies on the circle $x^2 + y^2 = 7225$. Find the equation of the tangent to the circle at this point.



Algebra › Functions › Function Notation (Foundation)

Topic 96 of 313 • id: functions_evaluate_F

Q1. $f(x) = 4x - 2$. Find the value of x for which $f(x) = 10$.

Algebra › Functions › Composite and Inverse Functions (Higher)

Topic 97 of 313 • id: functions_composite_inverse_H

Q1. $f(x) = -2x + 6$ and $g(x) = -5x + 1$. Find $fg(x)$.

Algebra › Functions › Evaluating Inverse Functions (Higher)

Topic 98 of 313 • id: functions_inverse_evaluate_H

Q1. $f(x) = 4x + 1$. Find $f^{-1}(0)$.

Algebra › Algebraic Fractions › Adding and Subtracting Algebraic Fractions (Higher)

Topic 99 of 313 • id: algebraic_fractions_add_subtract_H

Q1. Simplify $\frac{9}{x-4} - \frac{2}{x-1}$, giving your answer as a single fraction.

Algebra › Algebraic Fractions › Multiplying and Dividing Algebraic Fractions (Higher)

Topic 100 of 313 • id: algebraic_fractions_multiply_divide_H

Q1. Simplify $\frac{x^2 - 16}{x - 2} \times \frac{x - 2}{x - 4}$, giving your answer in its simplest form.

Algebra › Algebraic Fractions › Fractional Equations (Unknown in the Denominator) (Higher)

Topic 101 of 313 • id: fractional_equations_advanced_H

Solve the following equation, giving your answers to 2 decimal places where necessary.

Q1. $\frac{8}{x-1} = \frac{x+2}{5}$

Algebra › Simultaneous Equations › Common Coefficient (Foundation)

Topic 102 of 313 • id: simultaneous_common_coefficient_F

Solve the simultaneous equations:

Q1. $2x + 4y = -18$ and $4x + 4y = -8$

Algebra › Simultaneous Equations › Different Coefficients (Higher)

Topic 103 of 313 • id: simultaneous_different_coefficient_H

Solve the simultaneous equations:

Q1. $5x + 2y = 51$ and $4x + 4y = 48$

Algebra › Simultaneous Equations › Forming and Solving (Foundation)

Topic 104 of 313 • id: simultaneous_forming_and_solving_F

Q1. 5 tickets and 5 programmes cost £6.10 in total. 5 tickets and 2 programmes cost £4.21 in total. Form and solve a pair of simultaneous equations to find the cost of one ticket and one programme.

Algebra › Simultaneous Equations › Quadratic Simultaneous Equations (Higher)

Topic 105 of 313 • id: simultaneous_quadratic_H

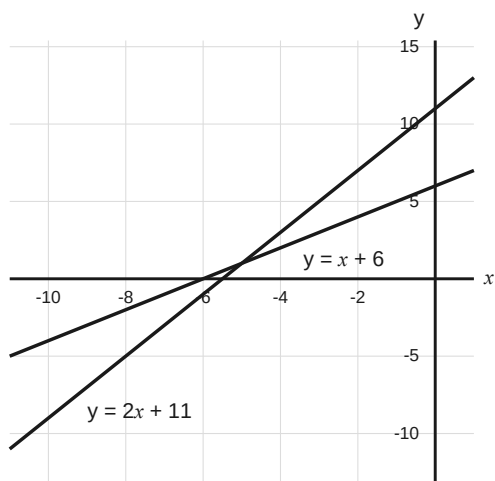
Solve the simultaneous equations:

Q1. $y = x^2 + 2x + 7$ and $y = -3x + 1$

Algebra › Simultaneous Equations › Solving Graphically (Foundation)

Topic 106 of 313 • id: simultaneous_graphically_F

Q1. The graphs of $y = x + 6$ and $y = 2x + 11$ are shown below. Use the graphs to write down the solution to the simultaneous equations.



Algebra › Inequalities › Solving Linear Inequalities (Foundation)

Topic 107 of 313 • id: solving_inequalities_F

Solve the inequality:

Q1. $2x + 9 > 5$

Algebra › Inequalities › Solving Linear Inequalities (Higher)

Topic 108 of 313 • id: solving_inequalities_H

Solve the inequality:

Q1. $-6(x + 9) < 6$

Algebra › Inequalities › Satisfying an Inequality (Foundation)

Topic 109 of 313 • id: satisfying_inequalities_F

Q1. Solve the inequality $-7 \leq 2x - 7 < 11$, then list the integer values of x that satisfy it.

Algebra › Inequalities › Satisfying an Inequality (Higher)

Topic 110 of 313 • id: satisfying_inequalities_H

Q1. x satisfies both $2x - 8 \geq 3$ and $2x - 7 \leq 7$. List the integer values of x that satisfy both.

Algebra › Inequalities › Solving Quadratic Inequalities (Higher)

Topic 111 of 313 • id: quadratic_inequalities_H

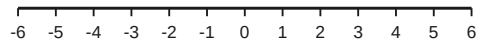
Solve the inequality:

Q1. $x^2 + 6x < 0$

Algebra › Inequalities › Inequalities on a Number Line (Foundation)

Topic 112 of 313 • id: inequalities_number_line_F

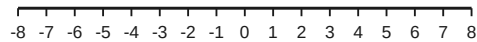
Q1. Draw the solution set of $x < 0$ on a number line.



Algebra › Inequalities › Compound Inequalities on a Number Line (Higher)

Topic 113 of 313 • id: inequalities_number_line_H

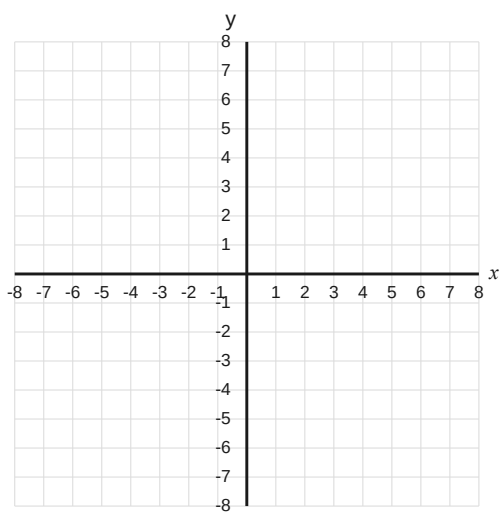
Q1. Draw the inequality/ies of $x \leq 2$ or $x > 6$ on a number line.



Algebra › Inequalities › Shading Inequality Regions (Higher)

Topic 114 of 313 • id: inequalities_region_H

Q1. On the grid, shade the region that satisfies both $y \geq \frac{x}{2} + 1$ and $y \geq 2x - 2$.



Algebra › Algebraic Proof › Algebraic Proof (Higher)

Topic 115 of 313 • id: algebraic_proof_H

Q1. Prove algebraically that $(n + 1)^3 - n^3 - 1$ is always a multiple of 6.

Algebra › Sequences › Next Term (Foundation)

Topic 116 of 313 • id: sequences_next_term_F

Find the next term in the sequence:

Q1. 6, 10, 14, 18, 22, ...

Algebra › Sequences › Term-to-Term Rule (Foundation)

Topic 117 of 313 • id: sequences_term_to_term_rule_F

Q1. Here are the first four terms of a sequence: -1, -3, -9, -27. Describe the term-to-term rule, and find the next term.

Algebra › Sequences › nth Term of a Linear Sequence (Foundation)

Topic 118 of 313 • id: sequences_nth_term_F

Q1. Here are the first four terms of a sequence: 8, 13, 18, 23.
Find an expression for the n th term.

Algebra › Sequences › nth Term of a Quadratic Sequence (Higher)

Topic 119 of 313 • id: sequences_quadratic_nth_term_H

Q1. Here are the first four terms of a quadratic sequence: -5, -9, -15, -23.
Find an expression for the n th term.

Algebra › Sequences › Special Sequences (Foundation)

Topic 120 of 313 • id: special_sequences_F

Q1. The first 4 triangular numbers are: 1, 3, 6, 10. Find the 6th term of the sequence.

Algebra › Sequences › Fibonacci-Type and Geometric Sequences (Higher)

Topic 121 of 313 • id: special_sequences_H

Q1. The first 5 terms of a Fibonacci-type sequence are: 8, 10, 18, 28, 46. Each term (from the 3rd onwards) is found by adding the two terms before it. Find the next two terms of the sequence.

Algebra › Iteration › Iteration (Higher)

Topic 122 of 313 • id: iteration_H

Q1. The iterative formula $x_{n+1} = \frac{17 - x_n^2}{9}$ is used with $x_0 = -1$. Find the values of x_1 , x_2 and x_3 , giving each answer to 3 decimal places.

Algebra › Iteration › Trial and Improvement (Higher)

Topic 123 of 313 • id: trial_and_improvement_H

Q1. $f(x) = x^3 - 2x - 72$. Show that $f(x) = 0$ has a root between $x = 4$ and $x = 5$, then use trial and improvement to find this root correct to 1 decimal place.

Algebra › Kinematics (SUVAT) › Kinematics (SUVAT) (Higher)

Topic 124 of 313 • id: kinematics_suvat_H

Formulae

$$v = u + at$$

$$s = ut + \left(\frac{1}{2}\right)at^2$$

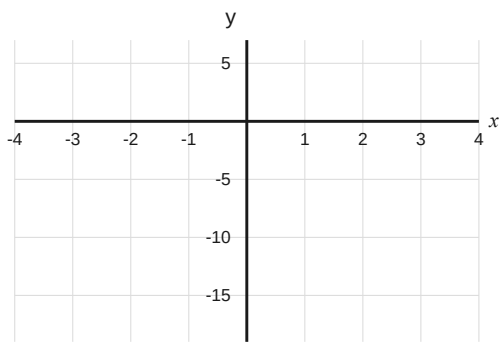
$$v^2 = u^2 + 2as$$

Q1. An object's velocity increases from 25 m/s (u) to 19 m/s (v) over a time (t) of 6 s at a constant acceleration. Find this acceleration (a).

Algebra › Plotting Graphs › Plotting Straight Line Graphs (Foundation)

Topic 125 of 313 • id: plot_straight_line_F

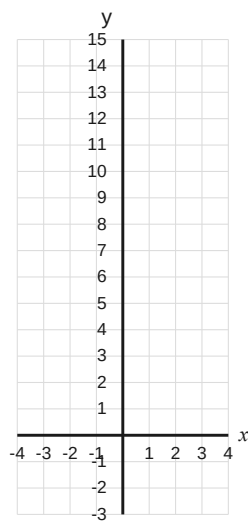
Q1. Complete a table of values for $y = 4x - 6$ for $x = -3$ to 3 , then plot the graph.



Algebra › Plotting Graphs › Plotting Quadratic Graphs (Foundation)

Topic 126 of 313 • id: plot_quadratic_F

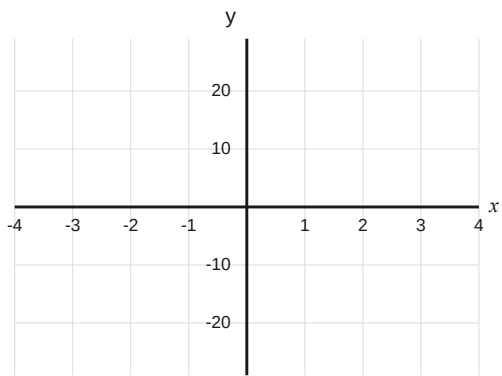
Q1. Complete a table of values for $y = x^2 - 2x - 1$ for $x = -3$ to 3 , then plot the graph.



Algebra › Plotting Graphs › Plotting Cubic Graphs (Higher)

Topic 127 of 313 • id: plot_cubic_H

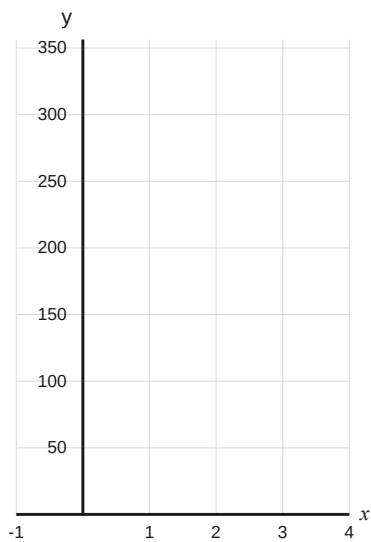
Q1. Complete a table of values for $y = -x^3$ for $x = -3$ to 3 , then plot the graph.



Algebra › Plotting Graphs › Plotting Exponential Graphs (Higher)

Topic 128 of 313 • id: plot_exponential_H

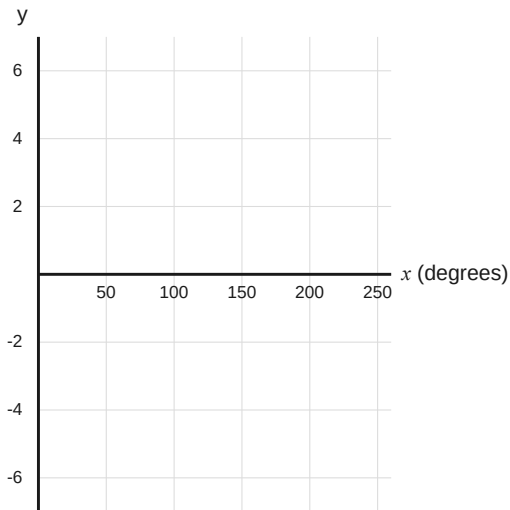
Q1. Complete a table of values for $y = 4 \times 3^x$ for $x = 0$ to 4, then plot the graph.



Algebra › Plotting Graphs › Plotting Trigonometric Graphs (Higher)

Topic 129 of 313 • id: trig_graph_H

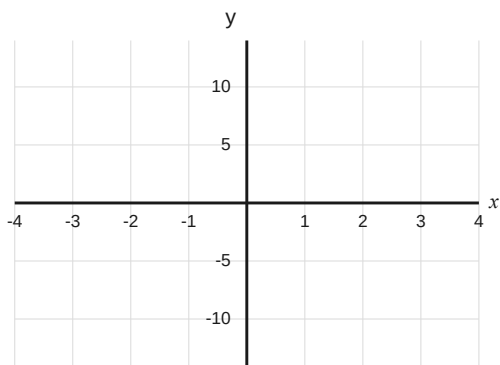
Q1. Complete a table of values for $y = -\tan(x^\circ)$ for $x = 100$ to 260 (in steps of 40°), then plot the graph.



Algebra › Plotting Graphs › Plotting Reciprocal Graphs (Higher)

Topic 130 of 313 • id: plot_reciprocal_H

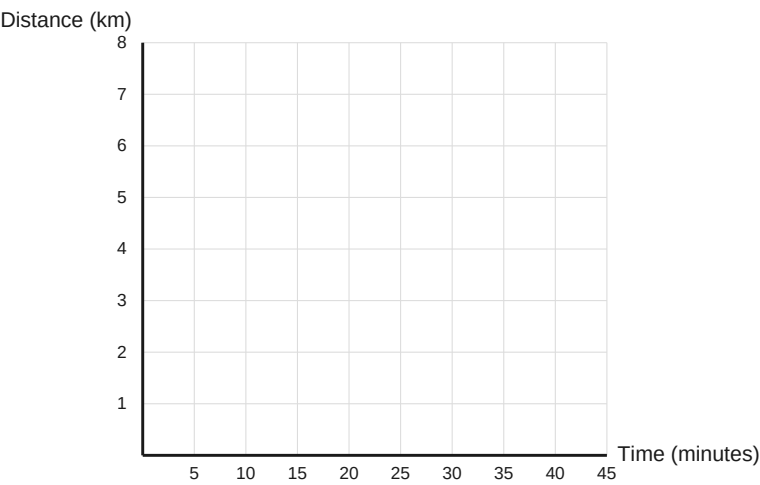
Q1. Complete a table of values for $y = \frac{12}{x}$ for $x = -4$ to 4 , then plot the graph.



Algebra › Plotting Graphs › Plotting Distance-Time Graphs (Foundation)

Topic 131 of 313 • id: plot_distance_time_F

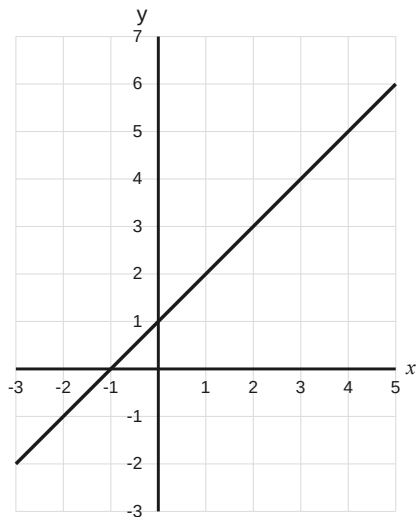
Q1. A cyclist leaves home at 0 minutes. They cycle at a constant speed, reaching a shop 6 km away after 15 minutes. They stay at the shop until 25 minutes, then cycle straight back home at a constant speed, arriving home at 40 minutes. Draw a distance-time graph to show this journey, and find the total distance travelled.



Algebra › Equation of a Line › Equation of a Line from a Graph (Foundation)

Topic 132 of 313 • id: line_equation_from_graph_F

Q1. The graph shows a straight line. Find the equation of the line in the form $y = mx + c$.



Algebra › Equation of a Line › Parallel Lines (Foundation)

Topic 133 of 313 • id: parallel_lines_equation_F

Q1. A line is parallel to $y = 3x + 2$ and passes through the point $(-1, 4)$. Find the equation of the line.

Algebra › Equation of a Line › Perpendicular Lines (Higher)

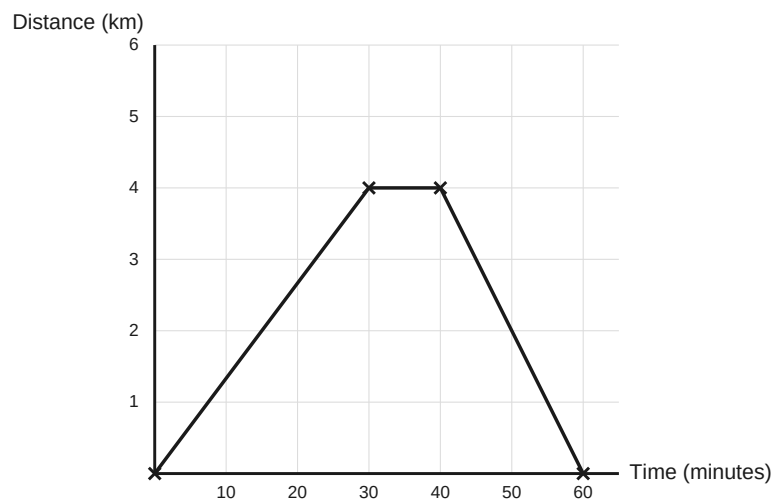
Topic 134 of 313 • id: perpendicular_lines_equation_H

Q1. A line is perpendicular to $y = -x + 1$ and passes through the point $(-2, -3)$. Find the equation of the line.

Algebra › Real-Life Graphs › Interpreting Distance-Time Graphs (Foundation)

Topic 135 of 313 • id: distance_time_interpret_F

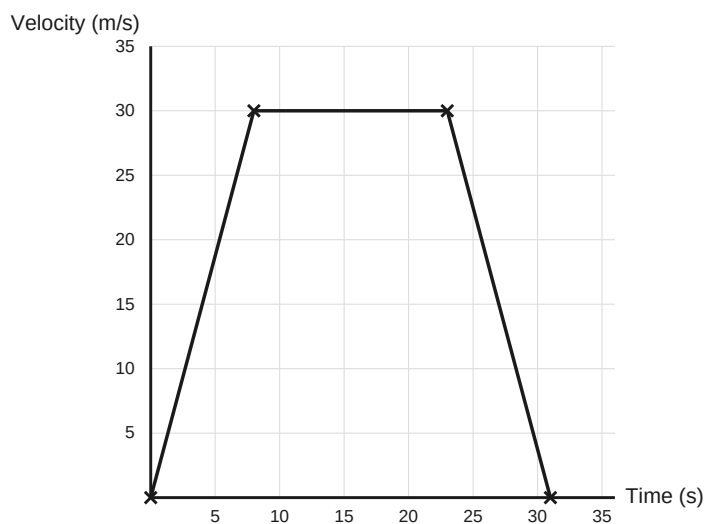
Q1. The distance-time graph shows a delivery van's journey. Find its speed during the outbound leg of the journey, in km/h.



Algebra › Real-Life Graphs › Interpreting Velocity-Time Graphs (Higher)

Topic 136 of 313 • id: velocity_time_interpret_H

Q1. The velocity-time graph shows a car's journey. Find the total distance travelled.



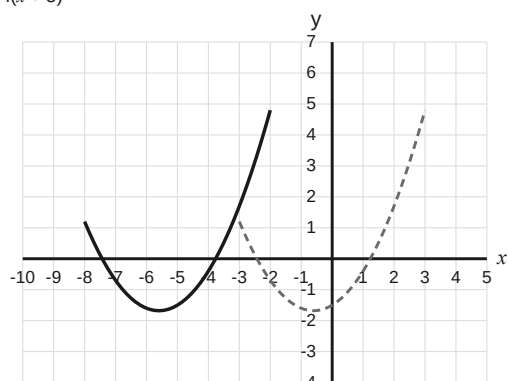
Algebra › Transformations of Graphs › Transformations of Graphs (Higher)

Topic 137 of 313 • id: graph_transformations_H

Q1. The graph of $y = f(x)$ is transformed to give the graph of $y = f(x + 5)$. Describe this transformation fully.

$$y = f(x)$$

$$y = f(x + 5)$$



Ratio & Proportion › Percentages › Percentage of an Amount (Foundation)

Topic 138 of 313 • id: percentage_of_amount_F

Q1. Find 20% of 200.

Ratio & Proportion › Percentages › Percentage Change (Foundation)

Topic 139 of 313 • id: percentage_change_F

Q1. Increase 240 by 20%.

Ratio & Proportion › Percentages › Finding a Percentage Change (Foundation)

Topic 140 of 313 • id: find_percentage_change_F

Q1. The price of an item changes from £20 to £23. Find the percentage increase.

Ratio & Proportion › Percentages › Percentage Increase/Decrease (Calculator) (Higher)

Topic 141 of 313 • id: percentage_increase_decrease_calculator_H

Q1. Using a calculator, decrease £56.00 by 9.75%. Give your answer to 2 decimal places.

Ratio & Proportion › Percentages › Reverse Percentage (Foundation)

Topic 142 of 313 • id: reverse_percentage_F

Q1. After a 20% increase, an item costs £384. Find the original price.

Ratio & Proportion › Percentages › Reverse Percentage (Higher)

Topic 143 of 313 • id: reverse_percentage_H

Q1. After a 12% decrease, an item costs £274.56. Find the original price.

Ratio & Proportion › Percentages › Compound Percentage Change (Higher)

Topic 144 of 313 • id: compound_percentage_H

Q1. A price of £260 decreases by 60%, then increases by 60%. Find the final price.

Ratio & Proportion › Percentages › Compound Percentage Change (Foundation)

Topic 145 of 313 • id: compound_percentage_F

Q1. A price of £40 increases by 25%, then increases by 50%. Find the final price.

Ratio & Proportion › Percentages › Simple Interest (Foundation)

Topic 146 of 313 • id: simple_interest_F

Q1. £2700 is invested at a simple interest rate of 5% per year for 1 years. Find the total amount in the account after 1 years.

Ratio & Proportion › Percentages › Mixed Percentages Practice (Higher)

Topic 147 of 313 • id: mixed_percentages_H

Q1. Decrease 380 by 20%.

Ratio & Proportion › Best Buys › Best Buys (Foundation)

Topic 148 of 313 • id: best_buys_F

Q1. A shop sells orange juice in different sizes: A: £12.50 for 1.25L; B: £25.20 for 1L; C: £22.80 for 1.5L. Which option is the better value for money? Show your working.

Ratio & Proportion › Best Buys › Best Buys (Non-Calculator) (Foundation)

Topic 149 of 313 • id: best_buys_noncalculator_F

Q1. A shop sells breakfast cereal in different sizes: A: £1.50 for 100g; B: £2.20 for 200g. Which option is the better value for money? Show your working.

Ratio & Proportion › Ratio › Share a Two-Part Ratio (Foundation)

Topic 150 of 313 • id: ratio_share_two_part_F

Q1. Share 130 in the ratio 4:9.

Ratio & Proportion › Ratio › Find a Missing Share (Foundation)

Topic 151 of 313 • id: ratio_find_missing_share_F

Q1. $p : q = 2 : 3$. $p = 18$. What is the value of q ?

Ratio & Proportion › Ratio › Share a Three-Part Ratio (Higher)

Topic 152 of 313 • id: ratio_share_three_part_H

Q1. Share 72 in the ratio 3:1:4.

Ratio & Proportion › Ratio › Share a Three-Part Ratio (Foundation)

Topic 153 of 313 • id: ratio_share_three_part_F

Q1. Share 70 in the ratio 8:5:1.

Ratio & Proportion › Ratio › Combine Ratios (Higher)

Topic 154 of 313 • id: ratio_combine_H

Q1. Given that $a:b = 5:9$ and $b:c = 8:2$, find $a:b:c$ in its simplest form.

Ratio & Proportion › Ratio › Ratio in the Form 1:n (Foundation)

Topic 155 of 313 • id: ratio_1_to_n_F

Q1. Write the ratio 11:4 in the form $n:1$.

Ratio & Proportion › Ratio › Ratio Given the Difference (Foundation)

Topic 156 of 313 • id: ratio_difference_F

Q1. $p : q = 4 : 7$.

$q - p = 15$.

Find p and q , and their total.

Ratio & Proportion › Ratio › Ratio Given the Difference (Three Parts) (Higher)

Topic 157 of 313 • id: ratio_difference_H

Q1. $p : q : r = 5 : 3 : 2$.

$p - r = 6$.

Find p , q and r , and their total.

Ratio & Proportion › Ratio › Ratio Problems with Algebra (Higher)

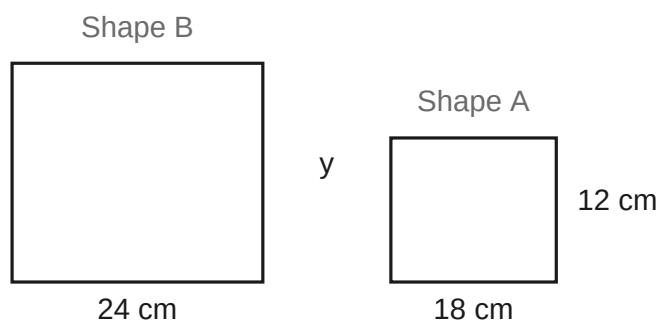
Topic 158 of 313 • id: ratio_to_equation_H

Q1. $(x + 1) : (3x - 6) = 2:3$. Find the value of x .

Ratio & Proportion › Ratio › Similar Shapes: Lengths (Foundation)

Topic 159 of 313 • id: ratio_shape_similar_F

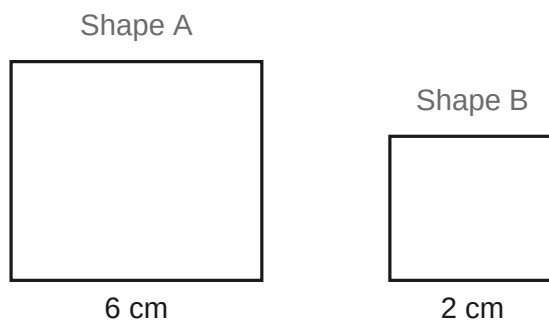
Q1. Shape A and Shape B are similar. Find the length of side y.



Ratio & Proportion › Ratio › Similar Shapes: Area and Volume (Higher)

Topic 160 of 313 • id: ratio_shape_similar_H

Q1. Shape A and Shape B are similar. The volume of shape A is 216 cm^3 . Find the volume of shape B.



Ratio & Proportion › Proportion › Direct Proportion (Foundation)

Topic 161 of 313 • id: direct_proportion_F

Q1. If 7 cm on a map represents 133 km in real life, how many km does 10 cm represent?

Ratio & Proportion › Proportion › Direct Proportion (Non-Calculator) (Foundation)

Topic 162 of 313 • id: direct_proportion_noncalculator_F

Q1. If 10 cm on a map represents 160 km in real life, how many km does 5 cm represent?

Ratio & Proportion › Proportion › Inverse Proportion (Foundation)

Topic 163 of 313 • id: inverse_proportion_F

Q1. It takes 8 machines 8 hours to complete a production run. Working at the same rate, how long would it take 2 machines?

Ratio & Proportion › Proportion › Inverse Proportion (Non-Calculator) (Foundation)

Topic 164 of 313 • id: inverse_proportion_noncalculator_F

Q1. It takes 6 workers 20 hours to build a wall. Working at the same rate, how long would it take 5 workers?

Ratio & Proportion › Proportion › Direct Proportion with Powers (Higher)

Topic 165 of 313 • id: algebraic_direct_proportion_H

Q1. y is directly proportional to x^2 . When $x = 6$, $y = 216$. Find the value of y when $x = 4$.

Ratio & Proportion › Proportion › Inverse Proportion with Powers (Higher)

Topic 166 of 313 • id: algebraic_inverse_proportion_H

Q1. y is inversely proportional to x^2 . When $x = 5$, $y = 4$. Find the value of y when $x = 2$.

Ratio & Proportion › Compound Measures › Speed, Distance and Time (Foundation)

Topic 167 of 313 • id: sdt_mixed_F

Q1. A delivery van travels 3432 m at an average speed of 26 m/s. Find how long the journey takes.

Ratio & Proportion › Compound Measures › Speed with Unit Conversions (Higher)

Topic 168 of 313 • id: speed_with_conversions_H

Q1. A train travels 2100 m in 75 seconds. Find its average speed in km/h, correct to 1 decimal place.

Ratio & Proportion › Compound Measures › Unit Conversions (Foundation)

Topic 169 of 313 • id: unit_conversions_F

Q1. Convert 431 cm to mm.

Ratio & Proportion › Compound Measures › Area and Volume Unit Conversions (Higher)

Topic 170 of 313 • id: unit_conversions_H

Q1. Convert 42 m^3 to cm^3 .

Ratio & Proportion › Compound Measures › Density (Foundation)

Topic 171 of 313 • id: density_F

Q1. A sculpture has a mass of 4368 kg and a density of 772 kg/m^3 . Find its volume, correct to 2 decimal places.

Ratio & Proportion › Compound Measures › Density (Multi-Step) (Higher)

Topic 172 of 313 • id: density_H

Q1. A stone has a mass of 16 kg and a volume of 1017 cm³. Find its density in g/cm³, correct to 2 decimal places.

Ratio & Proportion › Compound Measures › Pressure (Foundation)

Topic 173 of 313 • id: pressure_F

Q1. A concrete block exerts a force of 1284 N over a contact area of 26 m². Find the pressure, correct to 2 decimal places.

Ratio & Proportion › Compound Measures › Pressure (Multi-Step) (Higher)

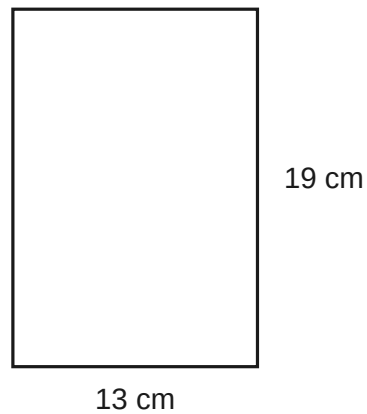
Topic 174 of 313 • id: pressure_H

Q1. A crate has a rectangular base measuring $29\text{ cm} \times 4\text{ cm}$ and exerts a force of 3499 N . Find the pressure it exerts, correct to 2 decimal places.

Geometry › Area & Perimeter › Rectangles (Foundation)

Topic 175 of 313 • id: area_rectangle_F

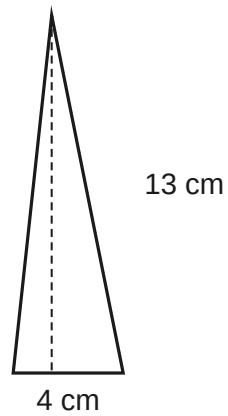
Q1. Find the perimeter of the following rectangle.



Geometry › Area & Perimeter › Triangles (Foundation)

Topic 176 of 313 • id: area_triangle_F

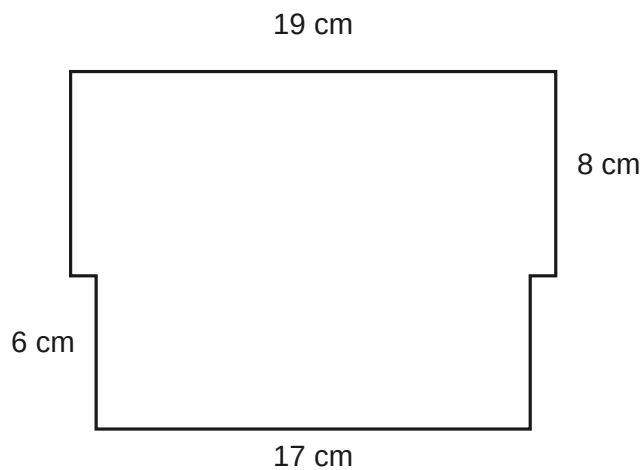
Q1. A triangle has base 4 cm and height 13 cm. Find its area.



Geometry › Area & Perimeter › Composite Rectangles (Foundation)

Topic 177 of 313 • id: area_composite_rectangles_F

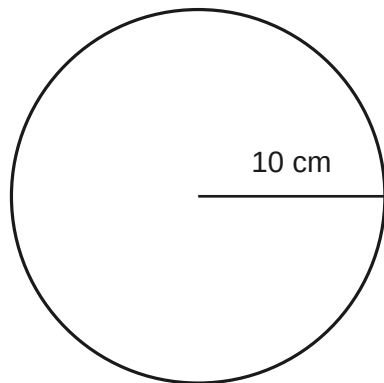
Q1. Find the area of this compound shape.



Geometry › Area & Perimeter › Circles (Calculator) (Foundation)

Topic 178 of 313 • id: area_circle_F

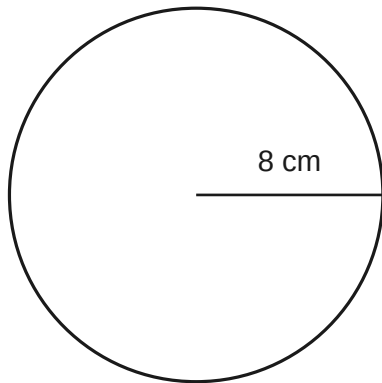
Q1. Find the circumference of the following circle, correct to 2 decimal places.



Geometry › Area & Perimeter › Circles (Higher)

Topic 179 of 313 • id: area_circle_H

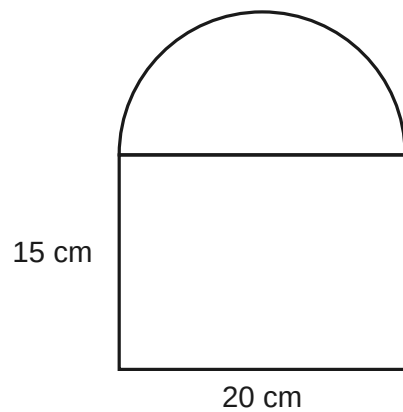
Q1. Find the circumference of the following circle, in terms of π .



Geometry › Area & Perimeter › Semicircle Compound Shapes (Calculator) (Foundation)

Topic 180 of 313 • id: area_semicircle_compound_F

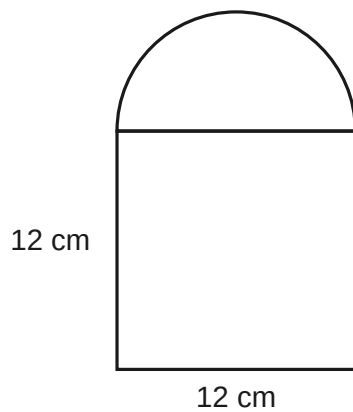
Q1. A shape is made from a rectangle 20 cm by 15 cm with a semicircle of diameter 20 cm attached to one side. Find the total area, correct to 3 s.f.



Geometry › Area & Perimeter › Semicircle Compound Shapes (Higher)

Topic 181 of 313 • id: area_semicircle_compound_H

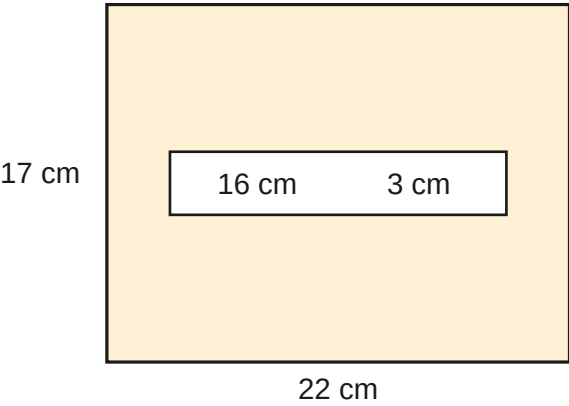
Q1. A shape is made from a rectangle 12 cm by 12 cm with a semicircle of diameter 12 cm attached to one side. Find the total area, giving your answer in terms of π .



Geometry › Area & Perimeter › Subtractive Compound Shapes (Higher)

Topic 182 of 313 • id: area_subtract_compound_H

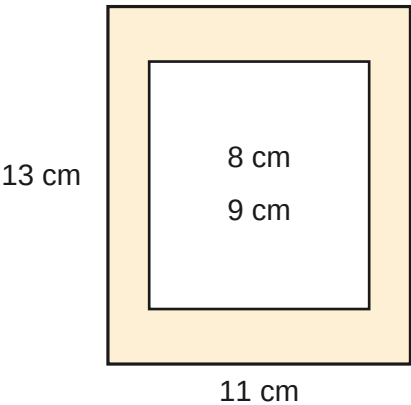
Q1. Find the shaded area.



Geometry › Area & Perimeter › Subtractive Compound Shapes (Foundation)

Topic 183 of 313 • id: area_subtract_compound_F

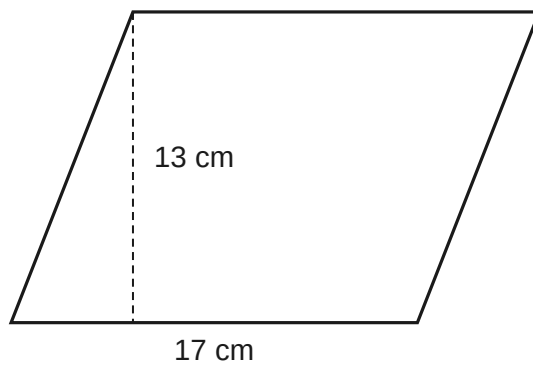
Q1. Find the shaded area.



Geometry › Area & Perimeter › Parallelograms (Foundation)

Topic 184 of 313 • id: area_parallelogram_F

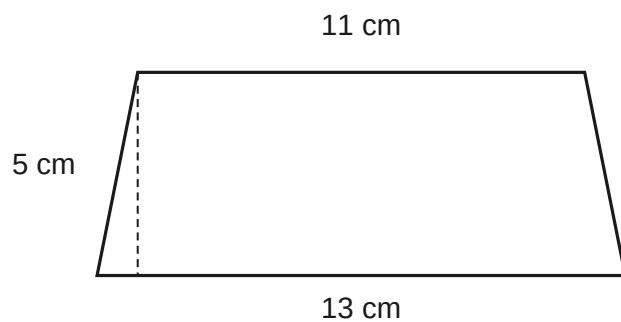
Q1. Find the area of the following parallelogram.



Geometry › Area & Perimeter › Trapeziums (Foundation)

Topic 185 of 313 • id: area_trapezium_F

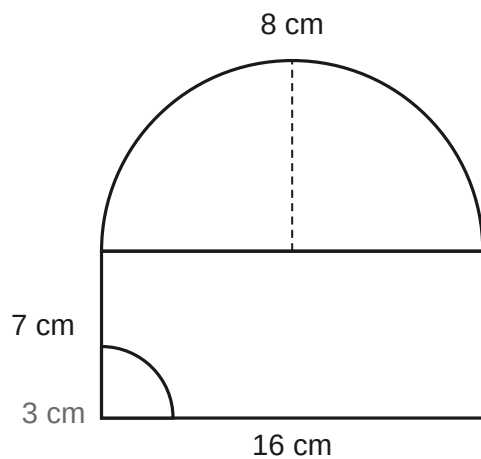
Q1. Find the area of the following trapezium.



Geometry › Area & Perimeter › Mixed Compound Shapes (Higher)

Topic 186 of 313 • id: area_mixed_compound_H

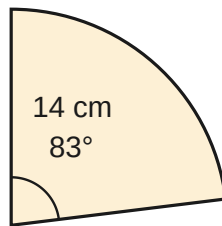
Q1. Find the total area of the following compound shape, correct to 1 decimal place.



Geometry › Area & Perimeter › Arc Length (Calculator) (Foundation)

Topic 187 of 313 • id: arc_length_F

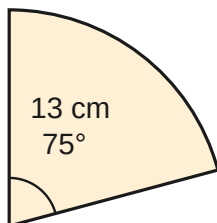
Q1. Find the arc length of the following sector, correct to 2 decimal places.



Geometry › Area & Perimeter › Arc Length (Higher)

Topic 188 of 313 • id: arc_length_H

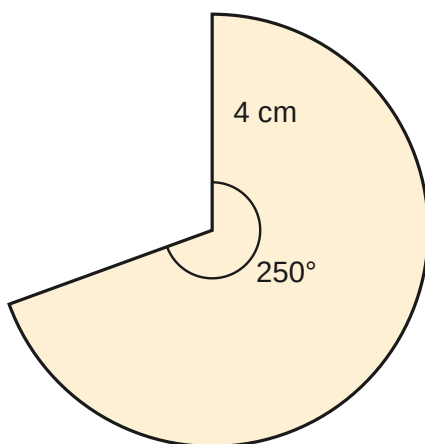
Q1. Find the exact arc length of the following sector, in terms of π .



Geometry › Area & Perimeter › Area of a Sector (Calculator) (Foundation)

Topic 189 of 313 • id: area_sector_F

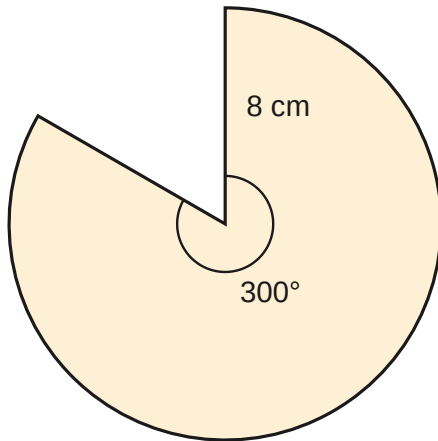
Q1. Find the area of the following sector, correct to 2 decimal places.



Geometry › Area & Perimeter › Area of a Sector (Higher)

Topic 190 of 313 • id: area_sector_H

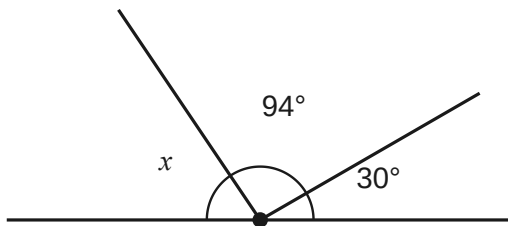
Q1. Find the exact area of the following sector, in terms of π .



Geometry › Angles › On a Straight Line (Foundation)

Topic 191 of 313 • id: angles_straight_line_F

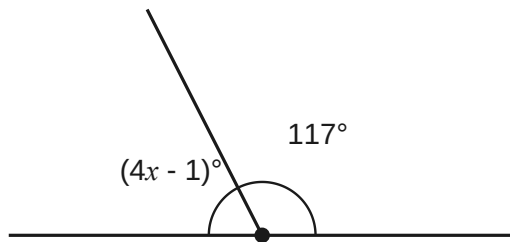
Q1. The angles 30° , 94° and x° lie on a straight line. Find x .



Geometry › Angles › On a Straight Line (Algebraic) (Higher)

Topic 192 of 313 • id: angles_straight_line_H

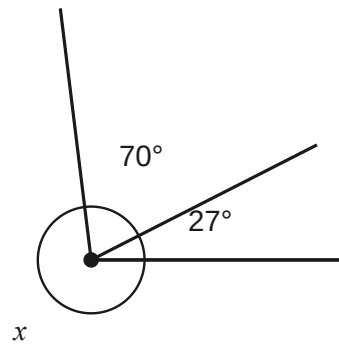
Q1. The angles 117° and $(4x - 1)^\circ$ lie on a straight line. Find x .



Geometry › Angles › Around a Point (Foundation)

Topic 193 of 313 • id: angles_around_point_F

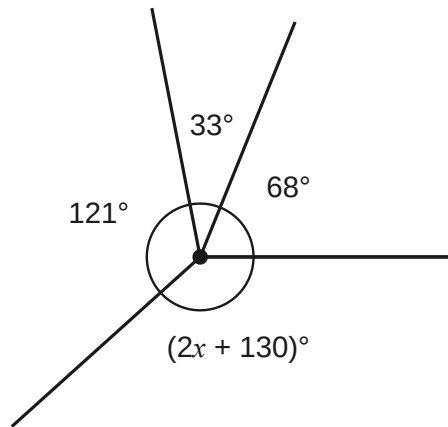
Q1. The angles 27° , 70° and x° are angles around a point. Find x .



Geometry › Angles › Around a Point (Algebraic) (Higher)

Topic 194 of 313 • id: angles_around_point_H

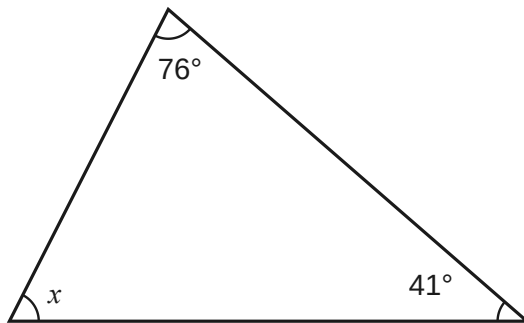
Q1. The angles 68° , 33° , 121° and $(2x + 130)^\circ$ are angles around a point. Find x .



Geometry › Angles › In a Triangle (Foundation)

Topic 195 of 313 • id: angles_triangle_F

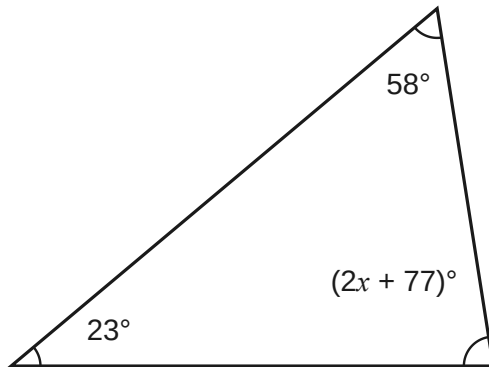
Q1. A triangle has angles 76° , 41° , and x° . Find x .



Geometry › Angles › In a Triangle (Algebraic) (Higher)

Topic 196 of 313 • id: angles_triangle_H

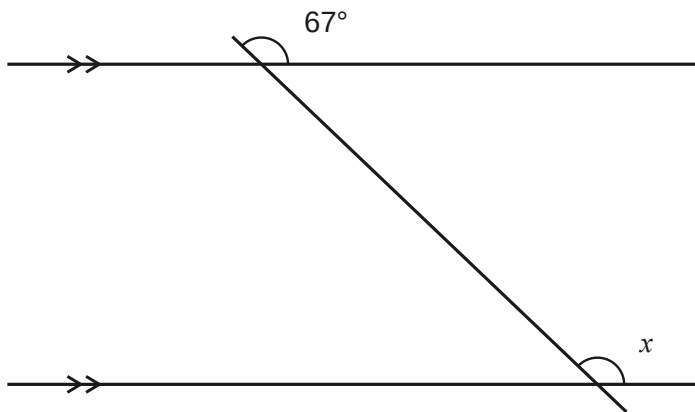
Q1. A triangle has angles 58° , 23° , and $(2x + 77)^\circ$. Find x .



Geometry › Angles › Parallel Lines (Foundation)

Topic 197 of 313 • id: angles_parallel_lines_F

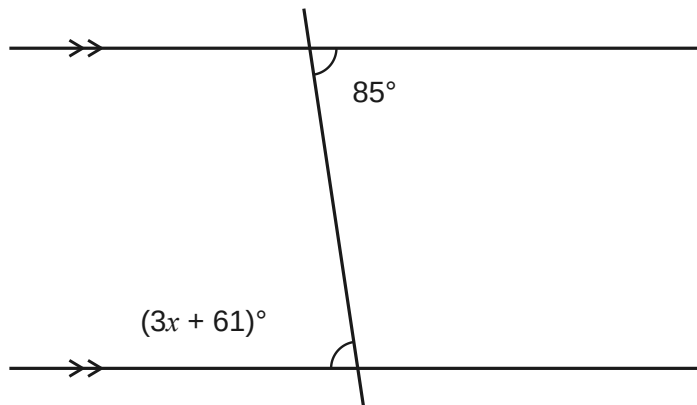
Q1. A line crosses two parallel lines. The angle 67° and the angle x° are corresponding angles. Find x .



Geometry › Angles › Parallel Lines (Higher)

Topic 198 of 313 • id: angles_parallel_lines_H

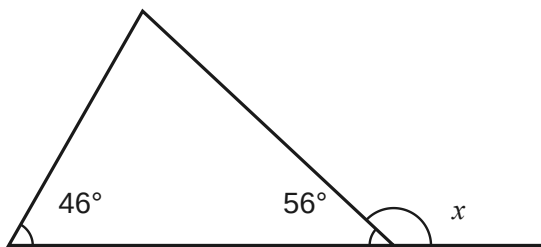
Q1. A line crosses two parallel lines. The angle 85° and the angle $(3x + 61)^\circ$ are alternate angles. Find x .



Geometry › Angles › Exterior Angle (Foundation)

Topic 199 of 313 • id: angles_exterior_F

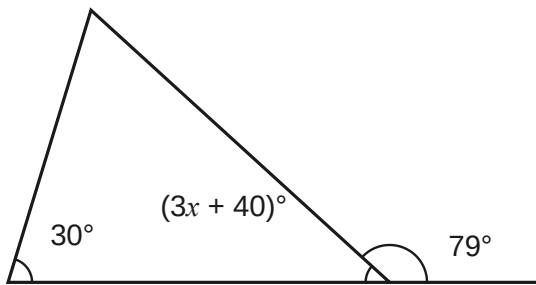
Q1. An exterior angle of a triangle is x° . The two remote interior angles are 46° and 56° . Find x .



Geometry › Angles › Exterior Angle (Higher)

Topic 200 of 313 • id: angles_exterior_H

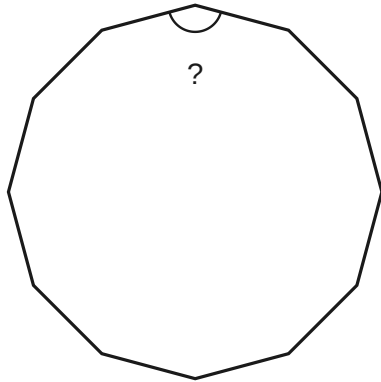
Q1. An exterior angle of a triangle is 79° . The two remote interior angles are 30° and $(3x + 40)^\circ$. Find x .



Geometry › Angles › Polygon Angles (Foundation)

Topic 201 of 313 • id: angles_polygon_interior_F

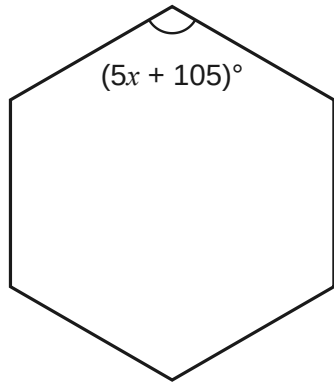
Q1. A regular polygon has 12 sides. Find the size of one interior angle.



Geometry › Angles › Polygon Interior Angles (Higher)

Topic 202 of 313 • id: angles_polygon_interior_H

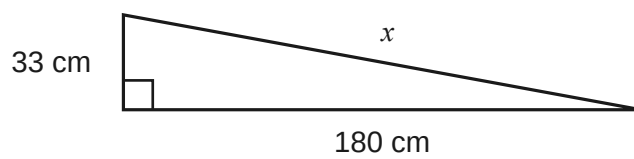
Q1. A polygon has 6 sides. 5 of its interior angles are each 114° , and the remaining interior angle is $(5x + 105)^\circ$. Find x .



Geometry › Pythagoras' Theorem › Hypotenuse (Triple) (Foundation)

Topic 203 of 313 • id: pythagoras_hypotenuse_triple_F

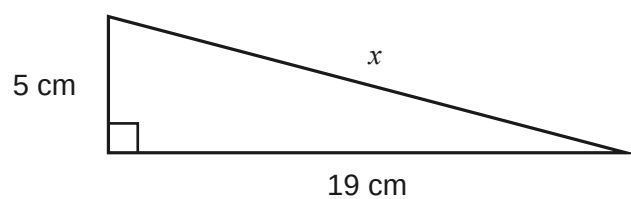
Q1. A right-angled triangle has sides 33 cm and 180 cm. Find the length of x .



Geometry › Pythagoras' Theorem › Hypotenuse (Decimal) (Foundation)

Topic 204 of 313 • id: pythagoras_hypotenuse_decimal_F

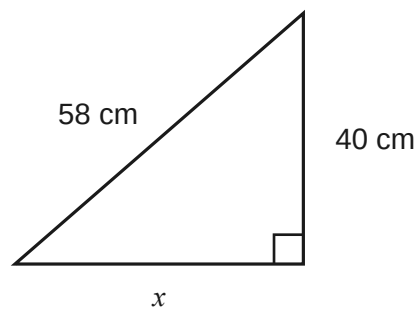
Q1. A right-angled triangle has sides 5 cm and 19 cm. Find the length of x , correct to 1 decimal place.



Geometry › Pythagoras' Theorem › Shorter Side (Foundation)

Topic 205 of 313 • id: pythagoras_shorter_leg_F

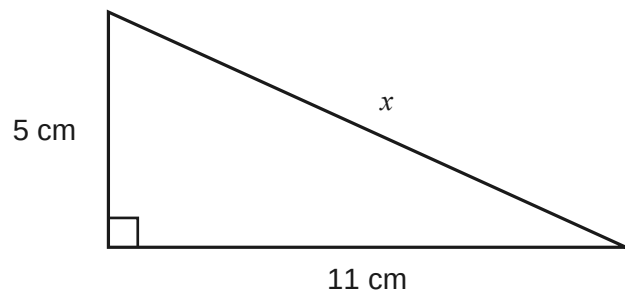
Q1. A right-angled triangle has hypotenuse 58 cm and one side 40 cm. Find the length of x .



Geometry › Pythagoras' Theorem › Exact Surd Hypotenuse (Higher)

Topic 206 of 313 • id: pythagoras_surd_hypotenuse_H

Q1. A right-angled triangle has sides 5 cm and 11 cm. Find the exact length of x , giving your answer as a simplified surd.



Geometry › Pythagoras' Theorem › Ladder Context (Higher)

Topic 207 of 313 • id: pythagoras_ladder_context_H

Q1. A ladder of length 15 m leans against a vertical wall with its base 8 m from the wall. Find the height the ladder reaches up the wall, giving your answer as a surd in its simplest form.

Geometry › Pythagoras' Theorem › Ladder Context (Foundation)

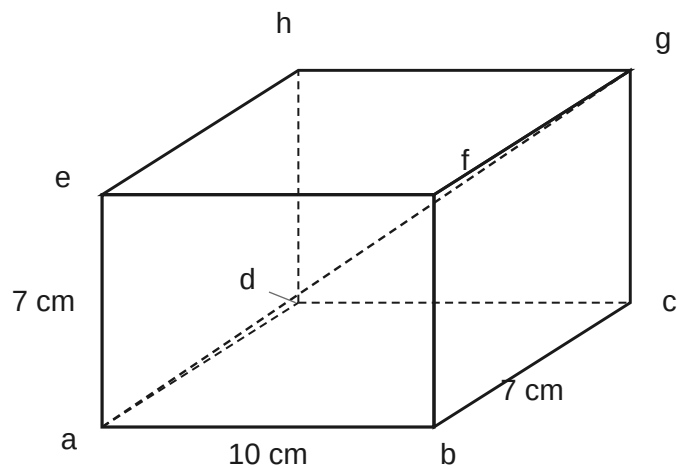
Topic 208 of 313 • id: pythagoras_ladder_context_F

Q1. A ladder of length 111 m leans against a vertical wall with its base 36 m from the wall. Find the height the ladder reaches up the wall.

Geometry › Pythagoras' Theorem › 3D Pythagoras (Higher)

Topic 209 of 313 • id: pythagoras_3d_H

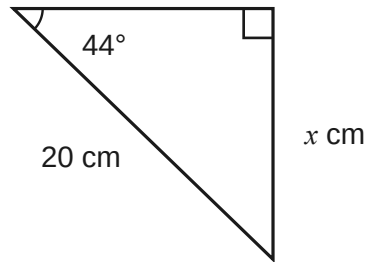
Q1. A cuboid has length 7 cm, width 10 cm and height 7 cm. Find the length of the diagonal ag that runs from one corner of the cuboid to the opposite corner, correct to 1 decimal place.



Geometry › Trigonometry › Missing Sides (Foundation)

Topic 210 of 313 • id: trig_missing_side_F

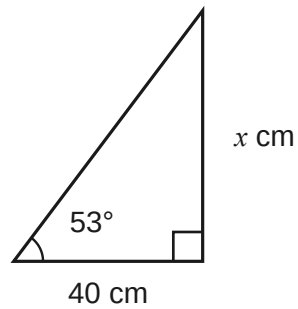
Q1. In the triangle shown, find the length of x , correct to 1 decimal place.



Geometry › Trigonometry › Missing Sides (Higher)

Topic 211 of 313 • id: trig_missing_side_H

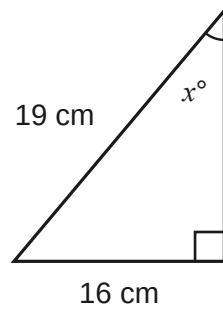
Q1. In the triangle shown, find the length of x , correct to 3 significant figures.



Geometry › Trigonometry › Missing Angles (Foundation)

Topic 212 of 313 • id: trig_missing_angle_F

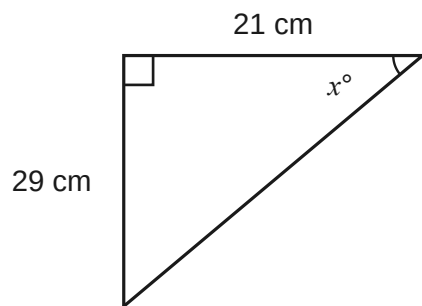
Q1. In the triangle shown, find the angle x , correct to 1 decimal place.



Geometry › Trigonometry › Missing Angles (Higher)

Topic 213 of 313 • id: trig_missing_angle_H

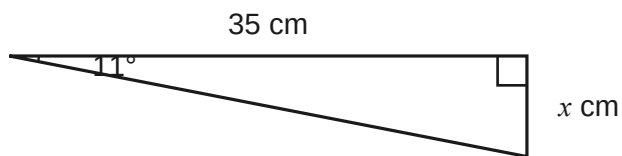
Q1. In the triangle shown, find the angle x , correct to 1 decimal place.



Geometry › Trigonometry › Mixed Sides and Angles (Higher)

Topic 214 of 313 • id: trig_mixed_H

Q1. In the triangle shown, find the length of x , correct to 1 decimal place.



Geometry › Trigonometry › Exact Trig Values (Higher)

Topic 215 of 313 • id: exact_trig_values_H

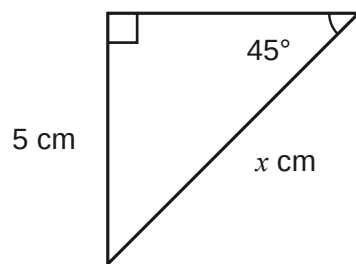
Write down the exact value of

Q1. $\cos(90^\circ)$.

Geometry › Trigonometry › Exact Trig Values in Triangles (Higher)

Topic 216 of 313 • id: exact_trig_values_triangles_H

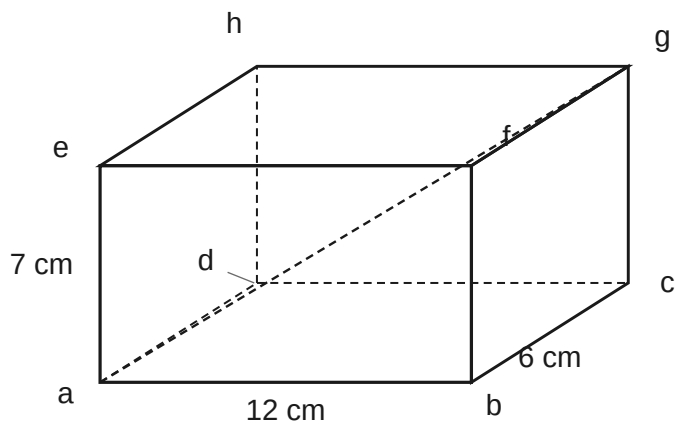
Q1. In the triangle shown, find the exact length of x . Give your answer in surd form where appropriate.



Geometry › Trigonometry › 3D Trigonometry (Higher)

Topic 217 of 313 • id: trig_3d_H

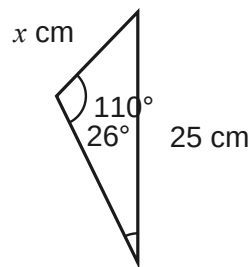
Q1. A cuboid has length 6 cm, width 12 cm and height 7 cm. Find the angle between the diagonal ag (from one corner to the opposite corner) and the base of the cuboid, correct to 1 decimal place.



Geometry › Sine Rule › Sine Rule (Higher)

Topic 218 of 313 • id: sine_rule_H

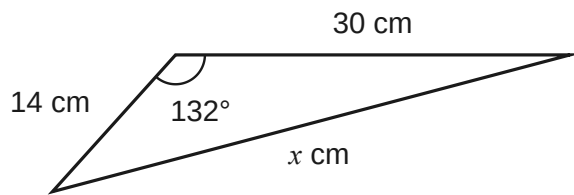
Q1. In triangle ABC, angle A = 110° , angle B = 26° , and side a = 25 cm. Find the length of side b, correct to 3 significant figures.



Geometry › Cosine Rule › Cosine Rule (Higher)

Topic 219 of 313 • id: cosine_rule_H

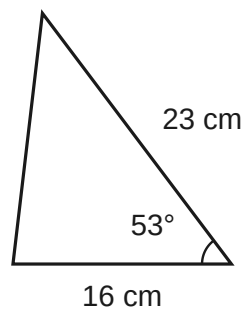
Q1. In triangle ABC, side $b = 14$ cm, side $c = 30$ cm, and the angle between them, angle $A = 132^\circ$. Find the length of side a , correct to 1 decimal place.



Geometry › Area of a Triangle › Area of a Triangle (Higher)

Topic 220 of 313 • id: triangle_area_sine_rule_H

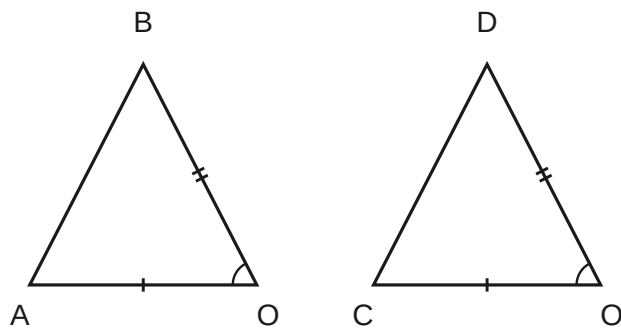
Q1. Triangle ABC has side $a = 23$ cm, side $b = 16$ cm, and the angle between them, angle $C = 53^\circ$. Find the area of the triangle, correct to 1 decimal place.



Geometry › Congruence Proof › Congruent Triangle Proof (Higher)

Topic 221 of 313 • id: congruent_triangle_proof_H

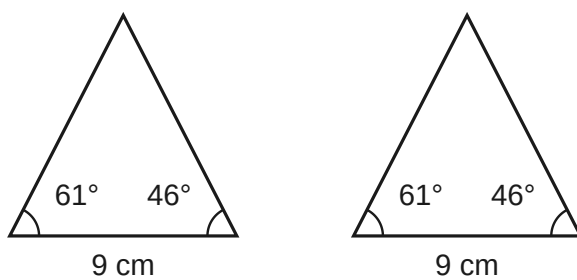
Q1. Lines AC and BD cross at point O, with $OA = OC$ and $OB = OD$. Prove that the two triangles are congruent.



Geometry › Congruence Proof › Congruent Triangle Proof (Foundation)

Topic 222 of 313 • id: congruent_triangle_proof_F

Q1. Shown below are two congruent triangles. Give a reason why they are congruent. A) SSS B) RHS C) ASA D) SAS



Geometry › Vectors › Vector Arithmetic (Foundation)

Topic 223 of 313 • id: vectors_arithmetic_F

Q1. $\mathbf{a} = \begin{pmatrix} 8 \\ 8 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$. Find $3\mathbf{a} - 4\mathbf{b}$.

Geometry › Vectors › Vector Arithmetic (Higher)

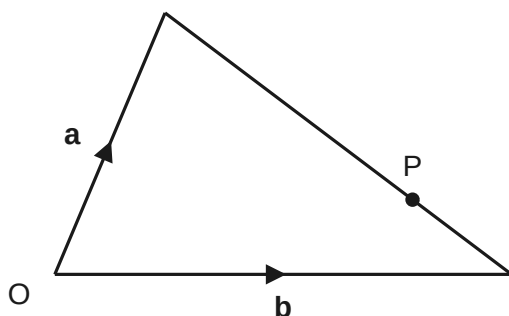
Topic 224 of 313 • id: vectors_arithmetic_H

Q1. $\mathbf{a} = \begin{pmatrix} 11 \\ -2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -9 \\ 2 \end{pmatrix}$. Find $-4\mathbf{a} + (-3\mathbf{b})$.

Geometry › Geometric Vectors › Vector Geometry (Higher)

Topic 225 of 313 • id: geometric_vectors_H

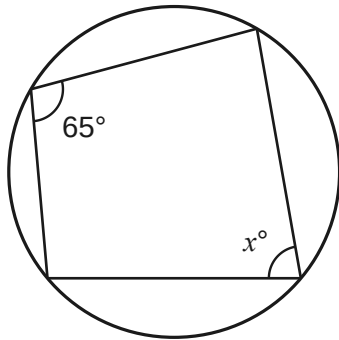
Q1. OAB is a triangle with $OA = \mathbf{a}$ and $OB = \mathbf{b}$. Point P lies on AB such that $AP:PB = 5:2$. Find the vector PB in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answer in its simplest form.



Geometry › Circle Theorems › Circle Theorems (Higher)

Topic 226 of 313 • id: circle_theorems_H

Q1. Find the size of angle x .



Geometry › 3D Shapes › Properties of 3D Shapes (Foundation)

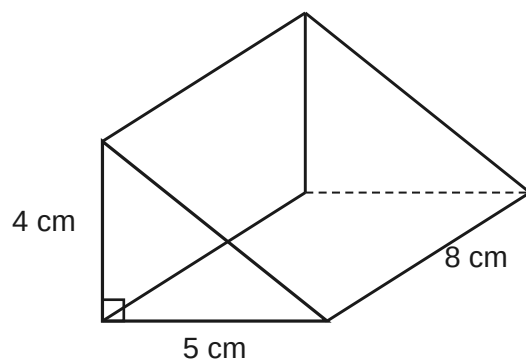
Topic 227 of 313 • id: properties_3d_shapes_F

Q1. How many faces does a cone have?

Geometry › 3D Shapes › Properties of 3D Shapes (with Diagram) (Foundation)

Topic 228 of 313 • id: properties_3d_shapes_diagram_F

Q1. How many vertices does the triangular prism shown have?



Geometry › 3D Shapes › Nets of 3D Shapes (Foundation)

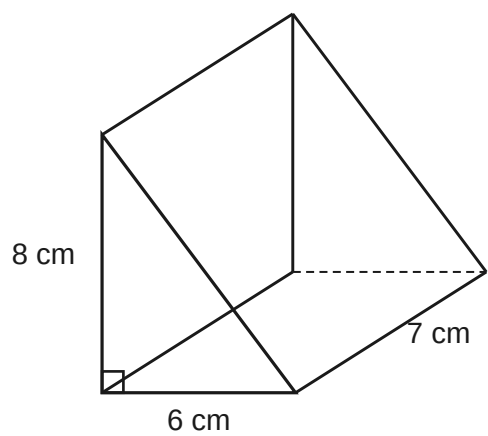
Topic 229 of 313 • id: nets_3d_shapes_F

Q1. What 2D shapes make up the net of a cuboid?

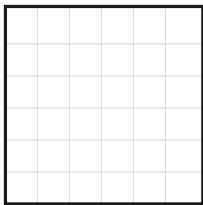
Geometry › 3D Shapes › Plans and Elevations (Foundation)

Topic 230 of 313 • id: plans_and_elevations_F

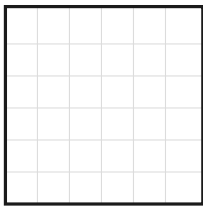
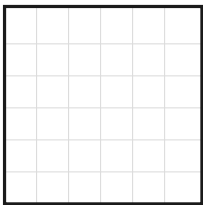
Q1. The diagram shows a solid with its dimensions labelled. Draw and label the front elevation, side elevation, and plan view of the solid.



Front elevation



Side elevation

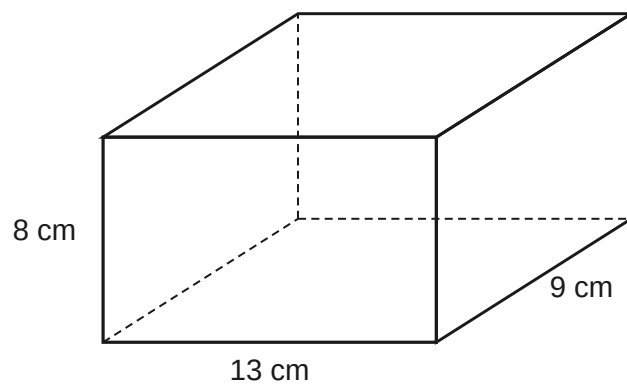


Plan view

Geometry › 3D Shapes › Volume & Surface Area of a Cuboid (Foundation)

Topic 231 of 313 • id: volume_surface_area_cuboid_F

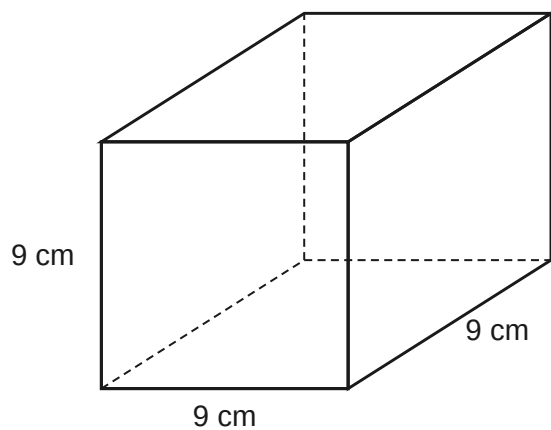
Q1. A cuboid has length 9 cm, width 13 cm and height 8 cm. Find its volume.



Geometry › 3D Shapes › Volume & Surface Area of a Cube (Foundation)

Topic 232 of 313 • id: volume_surface_area_cube_F

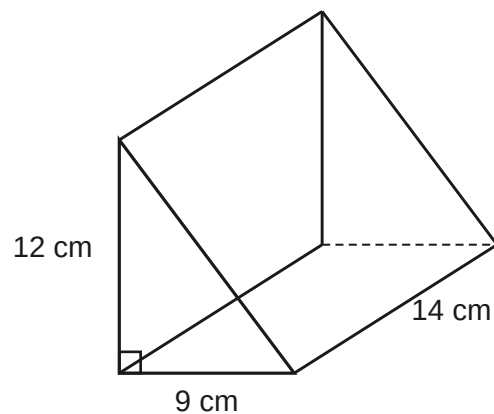
Q1. A cube has side length 9 cm. Find its surface area.



Geometry › 3D Shapes › Volume & Surface Area of a Triangular Prism (Foundation)

Topic 233 of 313 • id: volume_surface_area_triangular_prism_F

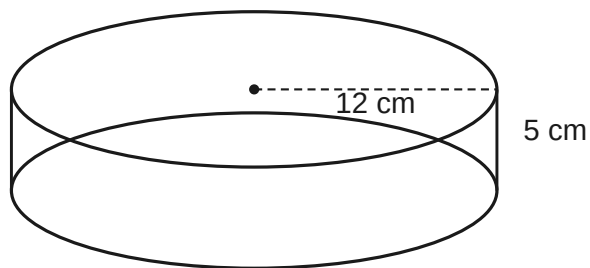
Q1. A triangular prism has a right-angled triangular cross-section with sides 9 cm and 12 cm (hypotenuse 15 cm), and length 14 cm. Find its volume.



Geometry › 3D Shapes › Volume & Surface Area of a Cylinder (Calculator) (Foundation)

Topic 234 of 313 • id: volume_surface_area_cylinder_F

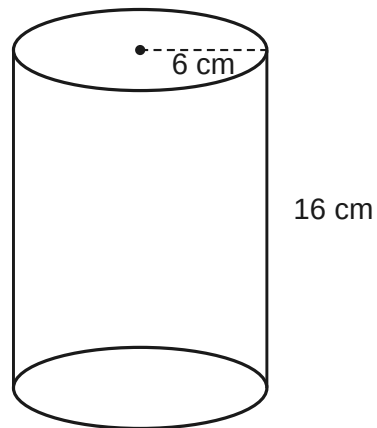
Q1. Find the total surface area of the cylinder, correct to 3 significant figures.



Geometry › 3D Shapes › Volume & Surface Area of a Cylinder (Higher)

Topic 235 of 313 • id: volume_surface_area_cylinder_H

Q1. Find the exact volume of the cylinder, in terms of π .



Geometry › 3D Shapes › Volume & Surface Area of a Cone (Higher)

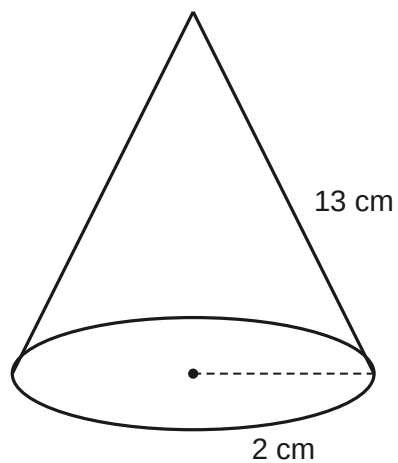
Topic 236 of 313 • id: volume_surface_area_cone_H

Formulae

Volume of a cone = $\left(\frac{1}{3}\right) \times \pi \times r^2 \times h$

Surface area of a cone = $\pi r(r + l)$, where l is the slant height

Q1. A cone has radius 2 cm and slant height 13 cm. Find its volume, correct to 2 decimal places.



Geometry › 3D Shapes › Volume & Surface Area of a Sphere (Higher)

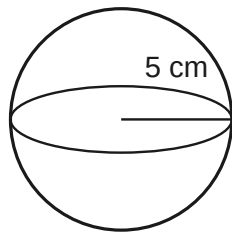
Topic 237 of 313 • id: volume_surface_area_sphere_H

Formulae

$$\text{Volume of a sphere} = \left(\frac{4}{3}\right) \times \pi \times r^3$$

$$\text{Surface area of a sphere} = 4 \times \pi \times r^2$$

Q1. A sphere has radius 5 cm. Find its volume, correct to 3 significant figures.



Geometry › 3D Shapes › Volume & Surface Area of a Pyramid (Higher)

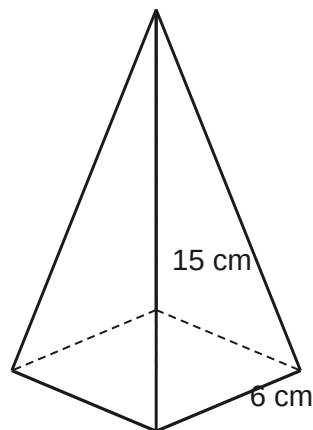
Topic 238 of 313 • id: volume_surface_area_pyramid_H

Formulae

Volume of a pyramid = $\left(\frac{1}{3}\right) \times \text{base}^2 \times \text{height}$

Surface area of a pyramid = $\text{base}^2 + 2 \times \text{base} \times l$, where l is the slant height of a face

Q1. A square-based pyramid has a base of side 6 cm and a perpendicular height of 15 cm. Find its total surface area, correct to 2 decimal places.



Geometry › 3D Shapes › Volume & Surface Area of a Frustum (Higher)

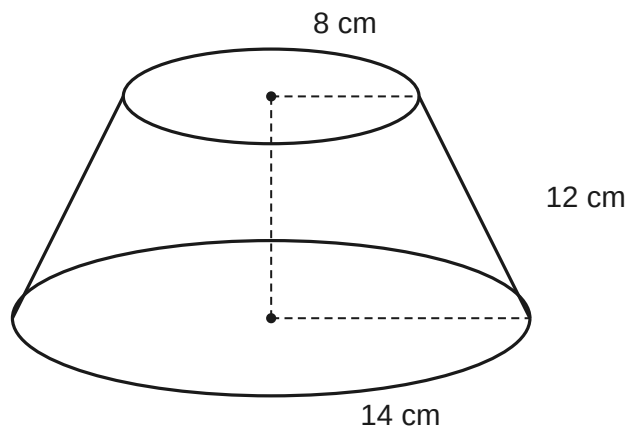
Topic 239 of 313 • id: frustum_volume_surface_area_H

Formulae

Volume of a frustum = $\left(\frac{1}{3}\right) \times \pi \times h \times (R^2 + Rr + r^2)$, where R and r are the two radii

Surface area of a frustum = $\pi(R + r)l + \pi R^2 + \pi r^2$, where l is the slant height

Q1. A frustum is formed from a cone with a bottom radius of 14 cm and a top radius of 8 cm, and a vertical height of 12 cm. Find its volume, correct to 2 decimal places.



Geometry › 3D Shapes › Compound 3D Shapes (Volume) (Higher)

Topic 240 of 313 • id: compound_3d_volume_H

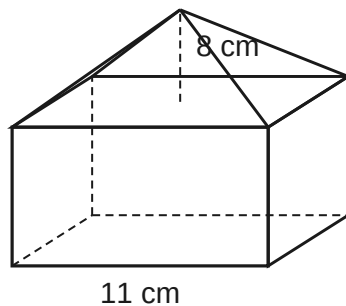
Formulae

Volume of a cone = $\left(\frac{1}{3}\right) \times \pi \times r^2 \times h$

Volume of a sphere = $\left(\frac{4}{3}\right) \times \pi \times r^3$ (a hemisphere is half this)

Volume of a pyramid = $\left(\frac{1}{3}\right) \times \text{base}^2 \times \text{height}$

Q1. A silo-shaped solid is made from a cuboid with a square base of side 11 cm and height 12 cm, with a square-based pyramid of height 8 cm on top (sharing the same square base). Find the total volume.



Geometry › 3D Shapes › Compound 3D Shapes (Surface Area) (Higher)

Topic 241 of 313 • id: compound_3d_surface_area_H

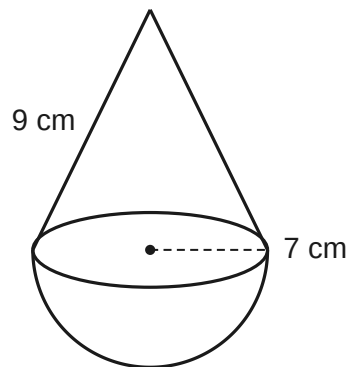
Formulae

Curved surface area of a cone = πrl , where l is the slant height

Curved surface area of a sphere = $4 \times \pi \times r^2$ (a hemisphere's curved part is half this)

Surface area of a pyramid's triangular faces = $2 \times \text{base} \times l$, where l is a face's slant height

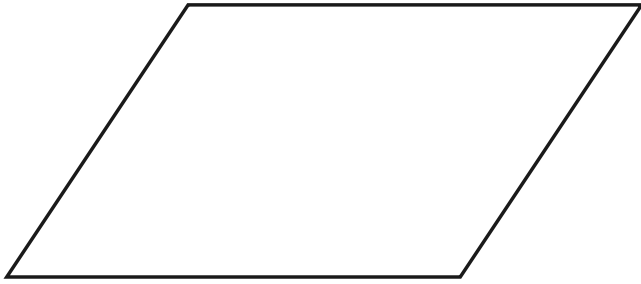
Q1. An ice-cream-shaped solid is made from a cone of radius 7 cm and height 9 cm, with a hemisphere of the same radius attached to its open end. Find the total surface area, correct to 1 decimal place.



Geometry › Symmetry › Line Symmetry (Foundation)

Topic 242 of 313 • id: symmetry_lines_F

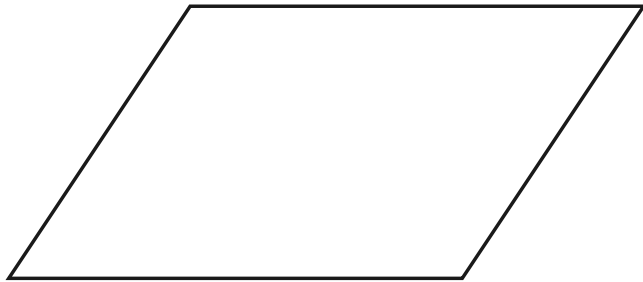
Q1. The diagram shows a parallelogram. How many lines of symmetry does it have?



Geometry › Symmetry › Rotational Symmetry (Foundation)

Topic 243 of 313 • id: symmetry_rotational_F

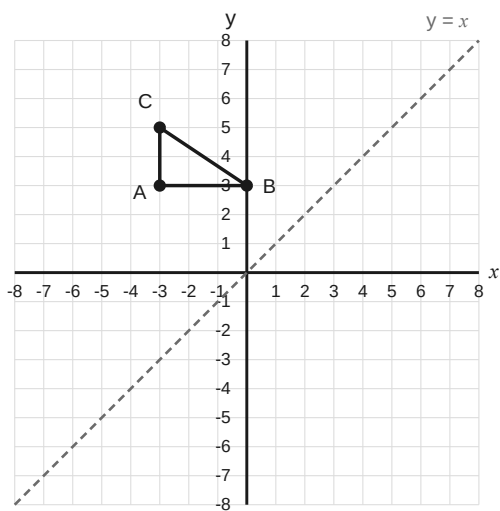
Q1. The diagram shows a parallelogram. State the order of rotational symmetry.



Geometry › Transformations › Reflection (Foundation)

Topic 244 of 313 • id: transform_reflect_complete_F

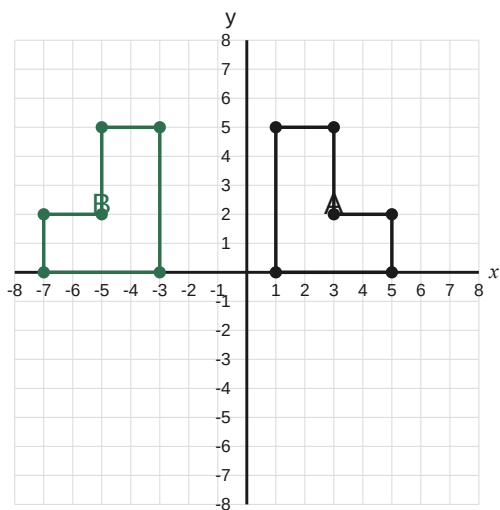
Q1. Reflect shape ABC in the line $y = x$.



Geometry › Transformations › Describe a Reflection (Foundation)

Topic 245 of 313 • id: transform_reflect_describe_F

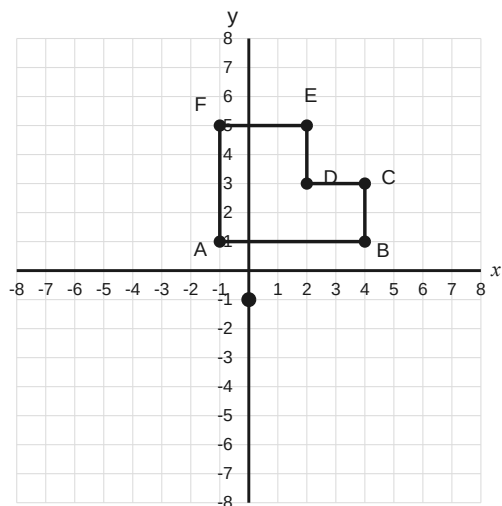
Q1. Describe fully the single transformation that maps shape A onto shape B, shown on the grid.



Geometry › Transformations › Rotation (Foundation)

Topic 246 of 313 • id: transform_rotate_complete_F

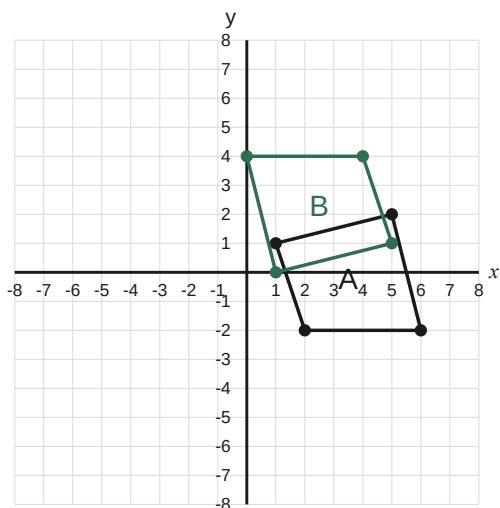
Q1. Rotate shape ABCDEF 180° about the centre $(0, -1)$.



Geometry › Transformations › Describe a Rotation (Higher)

Topic 247 of 313 • id: transform_rotate_describe_H

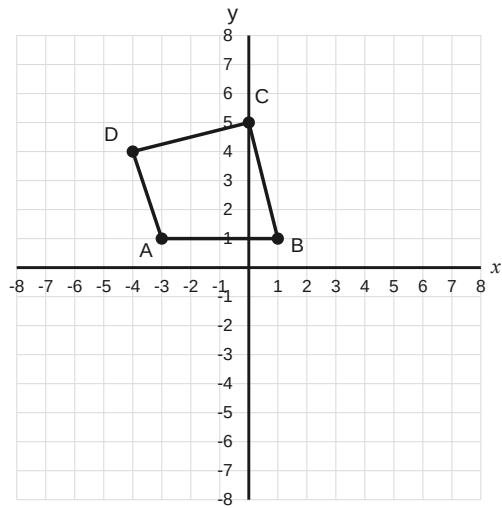
Q1. Describe fully the single transformation that maps shape A onto shape B, shown on the grid.



Geometry › Transformations › Translation (Foundation)

Topic 248 of 313 • id: transform_translate_complete_F

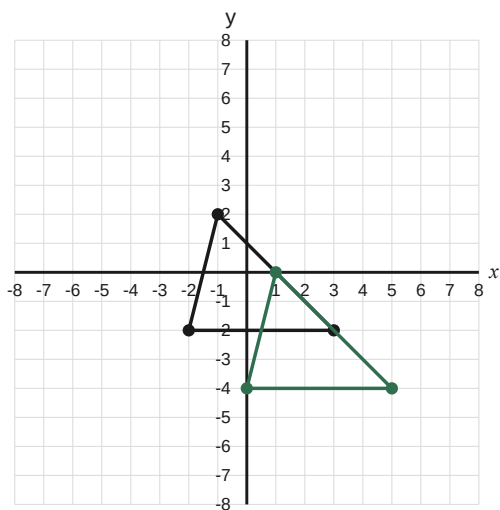
Q1. Translate shape ABCD by the vector $(4, 2)$.



Geometry › Transformations › Describe a Translation (Foundation)

Topic 249 of 313 • id: transform_translate_describe_F

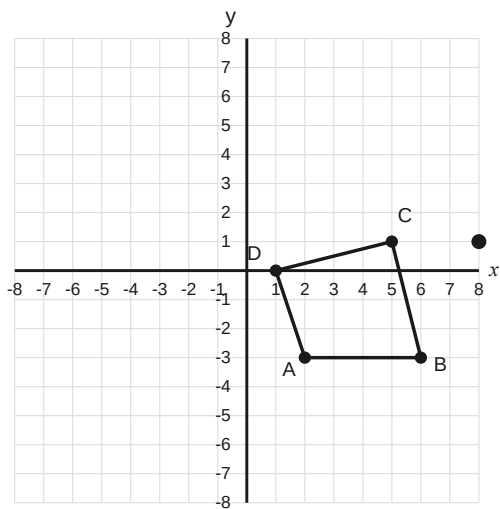
Q1. Describe fully the single transformation that maps shape A onto shape B, shown on the grid.



Geometry › Transformations › Enlargement (Foundation)

Topic 250 of 313 • id: transform_enlarge_complete_F

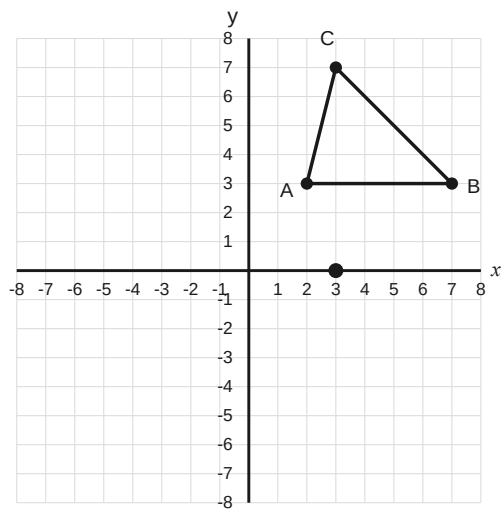
Q1. Enlarge shape ABCD by scale factor 2, centre (8, 1).



Geometry › Transformations › Enlargement (Higher)

Topic 251 of 313 • id: transform_enlarge_complete_H

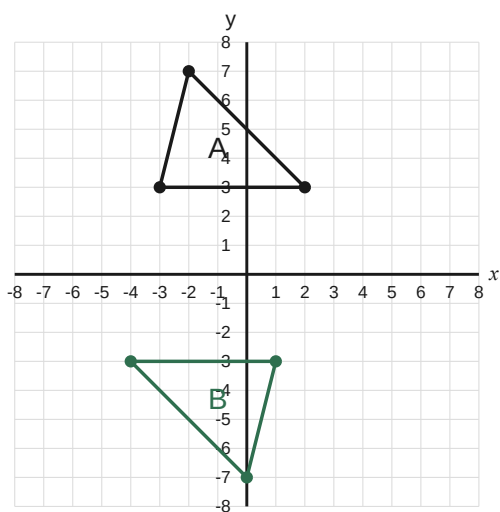
Q1. Enlarge shape ABC by scale factor -1 , centre $(3, 0)$.



Geometry › Transformations › Describe an Enlargement (Higher)

Topic 252 of 313 • id: transform_enlarge_describe_H

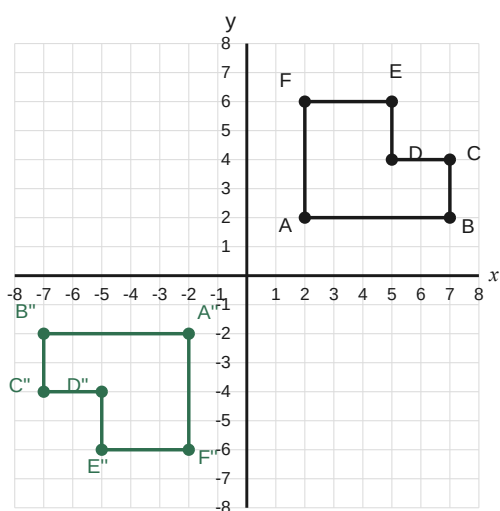
Q1. Describe fully the single transformation that maps shape A onto shape B, shown on the grid.



Geometry › Transformations › Combined Transformations (Higher)

Topic 253 of 313 • id: combined_transformations_H

Q1. Shape $ABCDEF$ is reflected in the y -axis to give shape $A'B'C'D'E'F'$. Shape $A'B'C'D'E'F'$ is then reflected in the x -axis to give shape $A''B''C''D''E''F''$. Describe fully the single transformation that maps $ABCDEF$ directly onto $A''B''C''D''E''F''$.



Geometry › Bearings › Bearings (Foundation)

Topic 254 of 313 • id: bearings_F

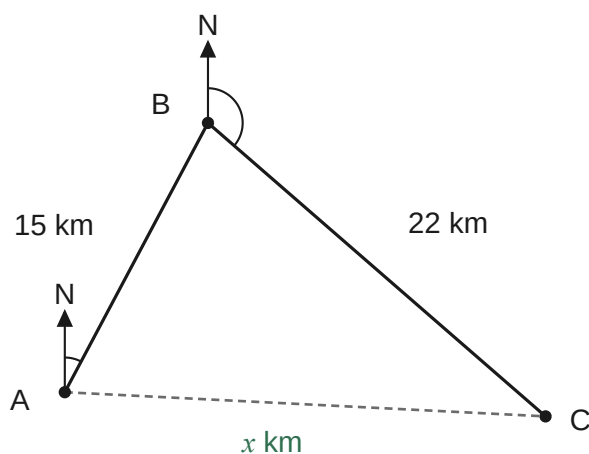
Q1. The diagram shows the bearing of B from A. Write down the bearing of B from A as a three-figure bearing.



Geometry › Bearings › Bearings (Higher)

Topic 255 of 313 • id: bearings_cosine_rule_H

Q1. A ship sails from port A on a bearing of 028° for 15 km to point B. It then changes course and sails on a bearing of 131° for 22 km to point C. Find the direct distance from A to C, correct to 3 significant figures.



Geometry › Map Scales and Scale Drawings › Map Scales and Scale Drawings (Foundation)

Topic 256 of 313 • id: map_scale_drawings_F

Q1. A map has a scale of 1 cm represents 5 km. Two towns are 65 km apart in real life. Find the distance between the towns on the map, in cm.

Geometry › Constructions › Constructing an Angle Bisector (Foundation)

Topic 257 of 313 • id: construction_angle_bisector_F

Q1. Angle $DYV = 76^\circ$. Describe, step by step, how to construct the bisector of angle DYV using only a ruler and a pair of compasses.

Geometry › Constructions › Constructing a Perpendicular Bisector (Foundation)

Topic 258 of 313 • id: construction_perpendicular_bisector_F

Q1. Line segment AK has length 9 cm. Describe, step by step, how to construct the perpendicular bisector of AK using only a ruler and a pair of compasses.

Geometry › Constructions › Constructing a Perpendicular From/At a Point (Foundation)

Topic 259 of 313 • id: construction_perpendicular_from_point_F

Q1. Point L lies on line segment RQ (length 5 cm). Describe, step by step, how to construct the perpendicular to RQ at L using only a ruler and a pair of compasses.

Geometry › Constructions › Constructing a Triangle (Foundation)

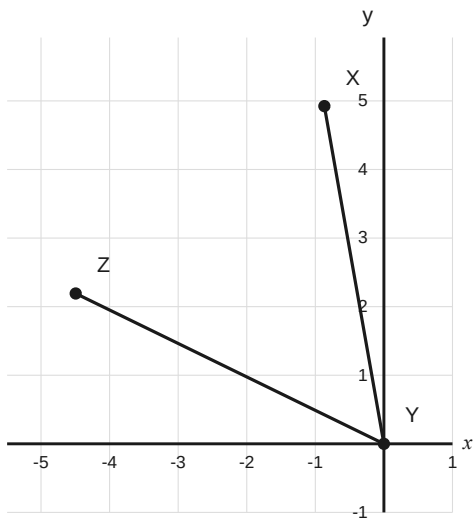
Topic 260 of 313 • id: construction_triangle_F

Q1. Triangle FPE has $FP = 10$ cm, $PE = 5$ cm, and $FE = 8$ cm. Describe, step by step, how to construct triangle FPE using only a ruler and a pair of compasses.

Geometry › Loci › Loci (Foundation)

Topic 261 of 313 • id: loci_constructions_F

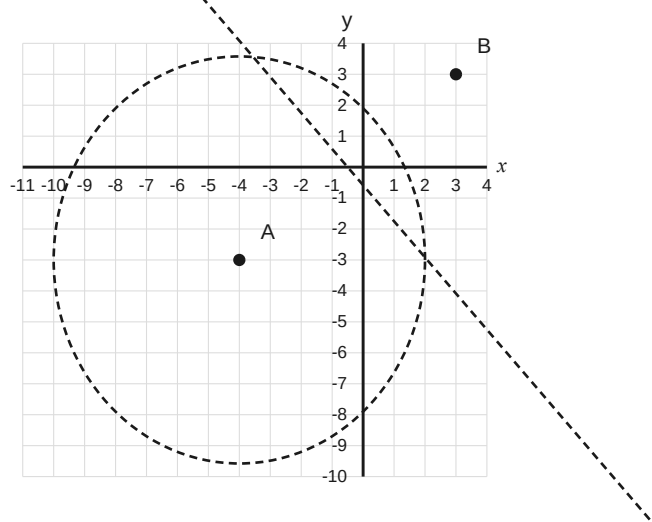
Q1. Two straight lines meet at Y, passing through X and Z as shown. Describe the locus of points that are equidistant from line XY and line ZY.



Geometry › Loci › Loci Regions (Higher)

Topic 262 of 313 • id: loci_regions_H

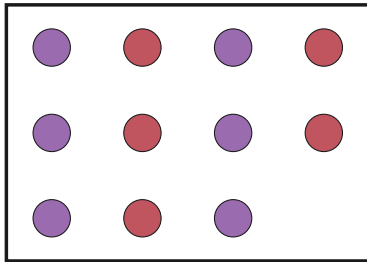
Q1. A and B are two fixed points (shown). Shade the region containing points that are both within 6 cm of A AND closer to A than to B.



Probability › Probability › Single Event (Foundation)

Topic 263 of 313 • id: probability_single_event_F

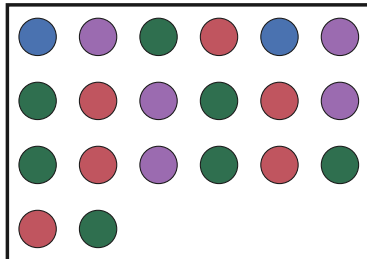
Q1. A bag contains 6 purple, 5 red counters. A counter is picked at random. Find the probability that it is red.



Probability › Probability › Complement (Foundation)

Topic 264 of 313 • id: probability_complement_F

Q1. A bag contains 2 blue, 5 purple, 7 green, 6 red counters. A counter is picked at random. Find the probability that it is NOT green.



Probability › Probability › Combined Dice (Higher)

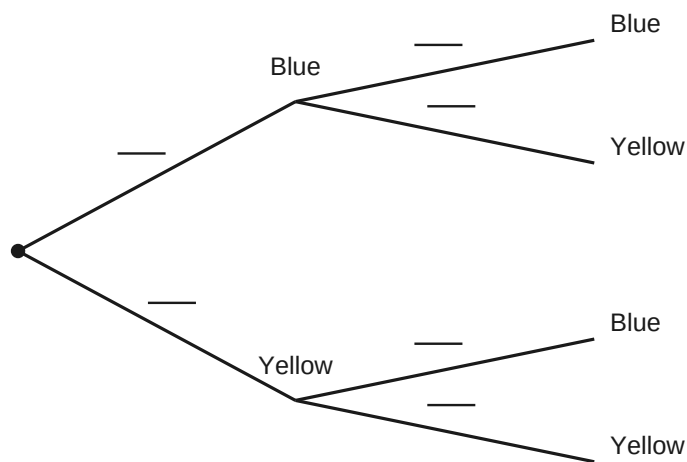
Topic 265 of 313 • id: probability_combined_dice_H

Q1. Two fair six-sided dice are rolled. Find the probability that at least one die shows 4.

Probability › Probability › Conditional (Without Replacement) (Higher)

Topic 266 of 313 • id: probability_conditional_H

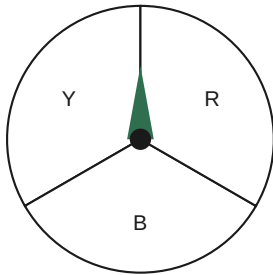
Q1. A bag contains 5 yellow and 6 blue counters. Two counters are picked at random without replacement. Find the probability that both counters are the same colour. Use the tree diagram to help you.



Probability › Probability › Listing Outcomes (Foundation)

Topic 267 of 313 • id: probability_listing_outcomes_F

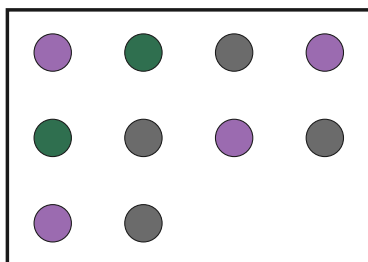
Q1. A coin is flipped and a 3-sided spinner (labelled Y, B, R) is spun. List all the possible outcomes.



Probability › Probability › AND / OR Rule (Foundation)

Topic 268 of 313 • id: probability_and_or_rule_F

Q1. A bag contains coloured counters. $P(\text{purple}) = \frac{4}{10}$. $P(\text{green}) = \frac{2}{10}$. A counter cannot be both colours. Find $P(\text{purple or green})$.



Probability › Probability › Expected Frequency (Foundation)

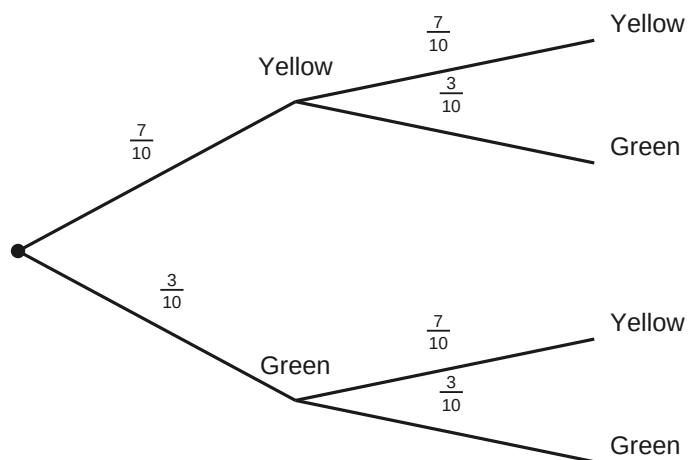
Topic 269 of 313 • id: probability_expectation_F

Q1. A biased coin has $P(\text{heads}) = \frac{5}{8}$. The coin is flipped 120 times. How many times would you expect to get heads?

Probability › Tree Diagrams › Interpreting Tree Diagrams (Independent Events) (Foundation)

Topic 270 of 313 • id: tree_diagram_independent_F

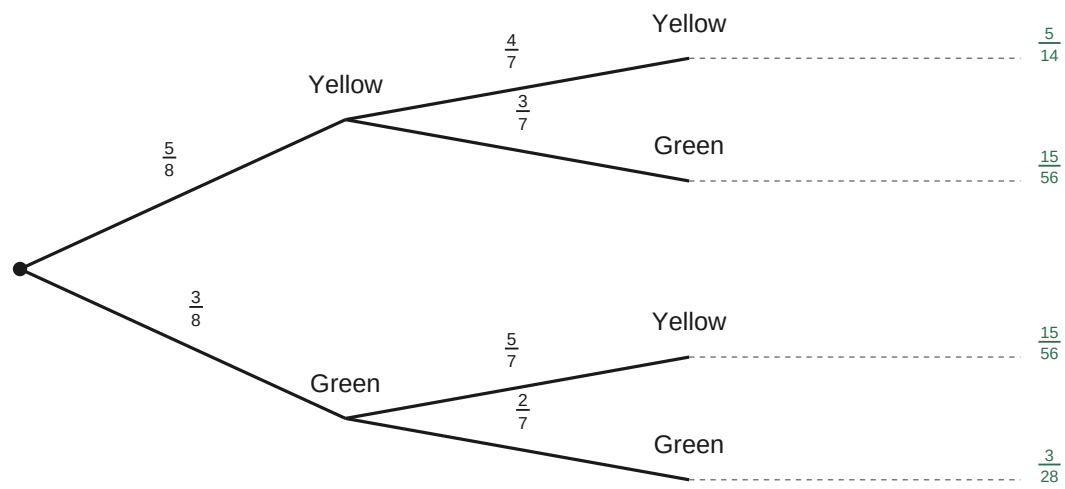
Q1. A bag contains 3 green and 7 yellow counters. A counter is picked at random, replaced, and then a second counter is picked at random. The tree diagram shows this information. Find the probability that both counters are the same colour.



Probability › Tree Diagrams › Interpreting Tree Diagrams (Dependent Events) (Higher)

Topic 271 of 313 • id: tree_diagram_dependent_H

Q1. A bag contains 3 green and 5 yellow counters. Two counters are picked at random, one after the other, without replacement. The tree diagram shows this information. Find the probability that the first counter is yellow and the second counter is yellow.



Probability › Tree Diagrams › Drawing Tree Diagrams (Foundation)

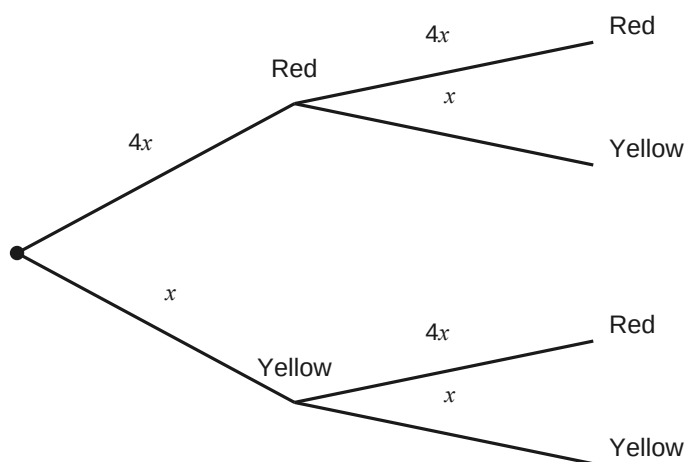
Topic 272 of 313 • id: tree_diagram_drawing_F

Q1. A bag contains 7 yellow and 8 green counters. A counter is picked at random, replaced, and then a second counter is picked at random. Draw a fully-labelled tree diagram to show this information, and use it to find the probability that the first counter is green and the second counter is yellow.

Probability › Tree Diagrams › Tree Diagrams with Algebraic Probabilities (Higher)

Topic 273 of 313 • id: tree_diagram_algebraic_H

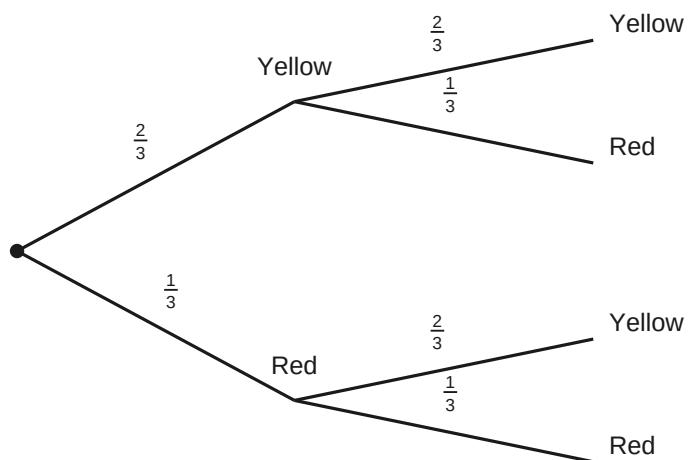
Q1. A biased spinner can only land on yellow or red. The probability that it lands on yellow is x , and the probability that it lands on red is $4x$. The spinner is spun twice. Find the probability that both spins land on red.



Probability › Tree Diagrams › Mixed Tree Diagrams (Higher)

Topic 274 of 313 • id: tree_diagram_mixed_H

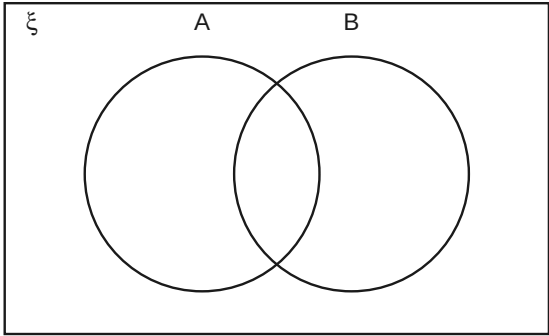
Q1. A bag contains 4 red and 8 yellow counters. A counter is picked at random, replaced, and then a second counter is picked at random. The tree diagram shows this information. Find the probability that both counters are the same colour.



Probability › Sets and Counting › Set Notation (Higher)

Topic 275 of 313 • id: set_notation_H

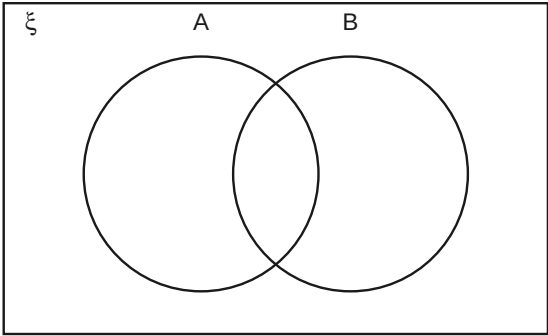
Q1. $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$, $A = \{1, 3, 4, 5, 7\}$, $B = \{2, 6, 8\}$. List the elements of $(A \cap B)'$.



Probability › Sets and Counting › Sets (Foundation)

Topic 276 of 313 • id: set_notation_F

Q1. $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 4, 5, 7, 9\}$, $B = \{1, 4, 5, 7, 8\}$. List the elements that are not in A.



Probability › Sets and Counting › Product Rule for Counting (Higher)

Topic 277 of 313 • id: product_rule_counting_H

Q1. A shop sells 4 different shirts and 3 different pairs of trousers. Use the product rule for counting to find how many different combinations are possible, choosing exactly one of each.

Probability › Tables and Diagrams › Relative Frequency (Foundation)

Topic 278 of 313 • id: relative_frequency_F

Q1. A coin is tested 40 times, and it lands on heads 24 times. Use this relative frequency to estimate how many times this would happen in 75 trials.

Probability › Tables and Diagrams › Two-Way Tables (Foundation)

Topic 279 of 313 • id: two_way_tables_F

Q1. The two-way table shows the number of boys and girls who prefer bus or walk. Find the missing values.

	Bus	Walk	Total
Boys		13	23
Girls	18		27
Total	28	22	50

Probability › Tables and Diagrams › Sample Space Diagrams (Foundation)

Topic 280 of 313 • id: sample_space_diagrams_F

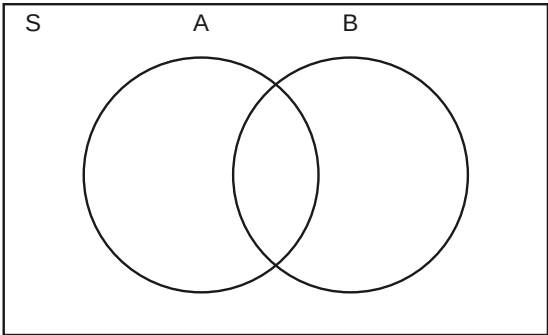
Q1. Two fair spinners, one numbered 1 to 4 and the other numbered 1 to 4, are spun together. Complete the sample space diagram showing the product of the two spinners, and use it to find the probability that the product is 9.

	1	2	3	4
1				
2				
3				
4				

Probability › Venn Diagrams › Venn Diagram Shading (Foundation)

Topic 281 of 313 • id: venn_diagram_shading_F

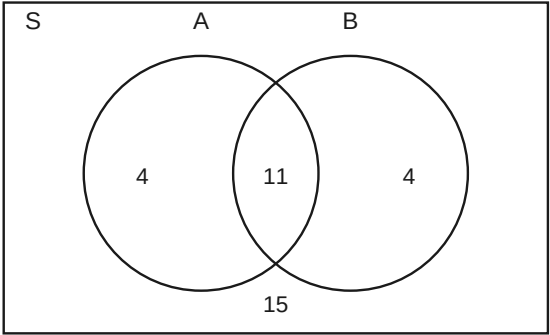
Q1. Shade the region that represents the elements in A but not in B.



Probability › Venn Diagrams › Probability from a Venn Diagram (Foundation)

Topic 282 of 313 • id: venn_diagram_probability_F

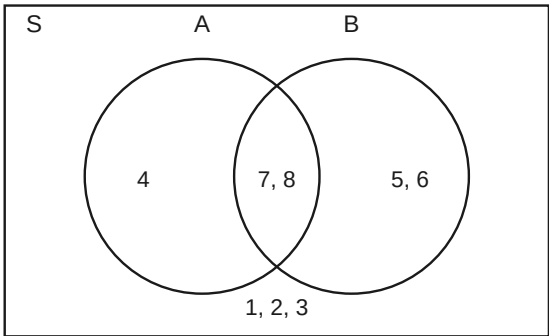
Q1. The Venn diagram shows the number of students in each region for sets A and B. Find $P(\text{A only})$.



Probability › Venn Diagrams › Venn Diagram Set Notation (Higher)

Topic 283 of 313 • id: venn_diagram_notation_H

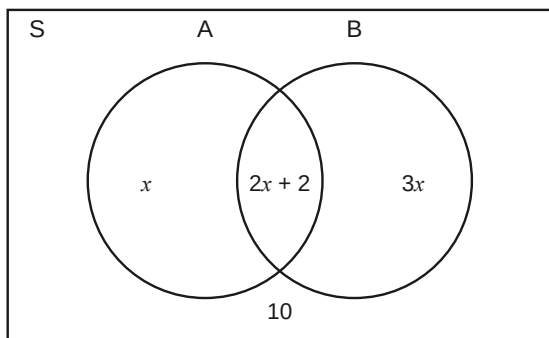
Q1. $\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$. The Venn diagram shows sets A and B. List the elements of $(A \cap B)'$.



Probability › Venn Diagrams › Algebraic Venn Diagrams (Higher)

Topic 284 of 313 • id: venn_diagram_algebra_H

Q1. The Venn diagram shows the number of elements in each region of sets A and B, given in terms of x . There are 54 elements in total. Find the value of x , and hence find the probability that a randomly chosen element is in A or B (or both).



Statistics › Averages from a List › Mean (Foundation)

Topic 285 of 313 • id: stats_mean_F

Find the mean of this data set:

Q1. 26, 17, 6, 14, 18.

Statistics › Averages from a List › Mode (Foundation)

Topic 286 of 313 • id: stats_mode_F

Find the mode of this data set:

Q1. 11, 3, 11, 12, 8, 3, 3, 7.

Statistics › Averages from a List › Median (Foundation)

Topic 287 of 313 • id: stats_median_F

Find the median of this data set:

Q1. 22, 12, 15, 8, 29.

Statistics › Averages from a List › Range (Foundation)

Topic 288 of 313 • id: stats_range_F

Find the range of this data set:

Q1. 2, 22, 29, 11, 13, 30.

Statistics › Averages from a List › Mean, Mode, Median & Range (Foundation)

Topic 289 of 313 • id: stats_averages_combined_F

Q1. For this data set, find (a) the mean (b) the mode (c) the median (d) the range: 9, 13, 18, 13, 13.

Statistics › Averages from a List › Interquartile Range (Higher)

Topic 290 of 313 • id: stats_interquartile_range_H

Q1. Find the interquartile range (IQR) of this data set: 33, 29, 32, 3, 19, 13, 21, 33, 6, 7, 11.

Statistics › Frequency Tables › Mean from a Frequency Table (Foundation)

Topic 291 of 313 • id: stats_mean_frequency_table_F

Q1. Find the mean number of pets.

	Frequency
0	10
1	2
2	3
3	5
4	5
5	8

Statistics › Frequency Tables › Mean from a Grouped Frequency Table (Higher)

Topic 292 of 313 • id: stats_mean_grouped_frequency_table_H

Q1. The table shows the times (in minutes) taken by a group of runners. Find an estimate of the mean time.

	Frequency
20-29	8
30-39	3
40-49	4
50-59	2

Statistics › Frequency Tables › Mean from a Grouped Frequency Table (Foundation)

Topic 293 of 313 • id: stats_mean_grouped_frequency_table_F

Q1. The table shows the times (in minutes) taken by a group of runners. Find an estimate of the mean time.

	Frequency
0-9	5
10-19	7
20-29	4

Statistics › Frequency Tables › Mode from a Frequency Table (Foundation)

Topic 294 of 313 • id: stats_mode_frequency_table_F

Q1. Find the modal number of pets.

	Frequency
0	3
1	2
2	4
3	6

Statistics › Frequency Tables › Median from a Frequency Table (Foundation)

Topic 295 of 313 • id: stats_median_frequency_table_F

Q1. Find the median number of pets.

	Frequency
0	5
1	6
2	8
3	8
4	9
5	8

Statistics › Frequency Tables › Range from a Frequency Table (Foundation)

Topic 296 of 313 • id: stats_range_frequency_table_F

Q1. Find the range of the number of pets.

	Frequency
2	2
3	8
4	3
5	10
6	2

Statistics › Working Backwards › Reverse Mean (Higher)

Topic 297 of 313 • id: stats_reverse_mean_H

Q1. The mean of four numbers is 16. Three of the numbers are 30, 17, 4. Find the missing number.

Statistics › Working Backwards › Reverse Mean (Foundation)

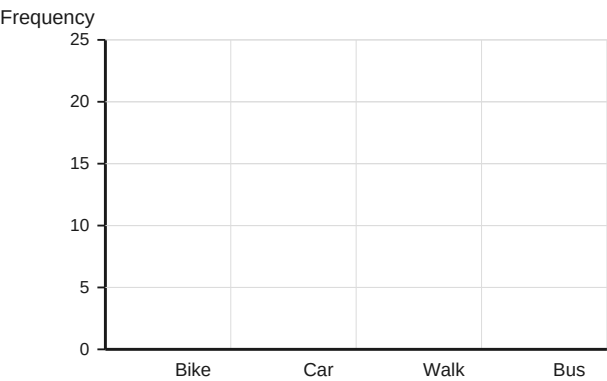
Topic 298 of 313 • id: stats_reverse_mean_F

Q1. The mean of four numbers is 7. Three of the numbers are 10, 2, 5. Find the missing number.

Statistics › Charts and Graphs › Constructing Bar Charts (Foundation)

Topic 299 of 313 • id: bar_chart_construct_F

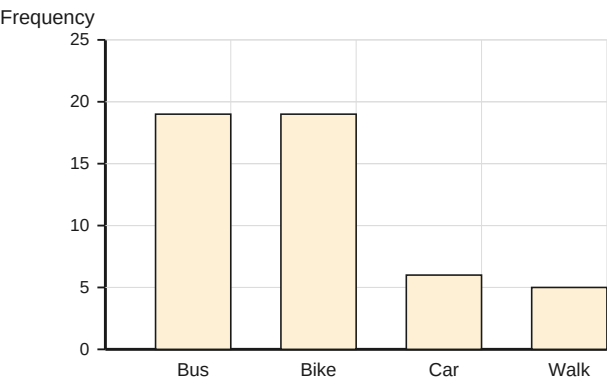
Q1. The table shows the mode of transport to school of a group of students: Bike: 7, Car: 19, Walk: 9, Bus: 6. Draw a bar chart to show this information.



Statistics › Charts and Graphs › Interpreting Bar Charts (Foundation)

Topic 300 of 313 • id: bar_chart_interpret_F

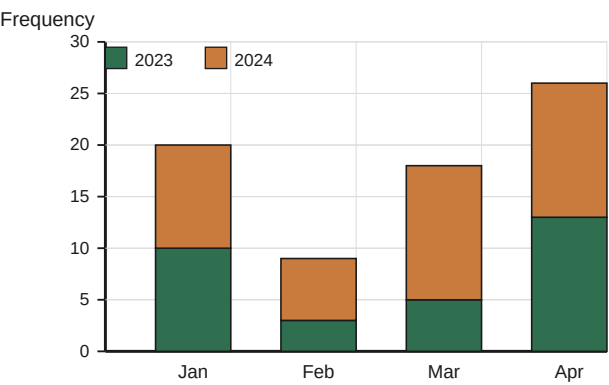
Q1. The bar chart shows the mode of transport to school of a group of students. How many chose Bus?



Statistics › Charts and Graphs › Composite Bar Charts (Foundation)

Topic 301 of 313 • id: composite_bar_chart_F

Q1. The composite bar chart shows the number of sales split by 2023 and 2024. Find the difference between 2023 and 2024 for Feb.



Statistics › Charts and Graphs › Constructing Pie Charts (Foundation)

Topic 302 of 313 • id: pie_chart_construct_F

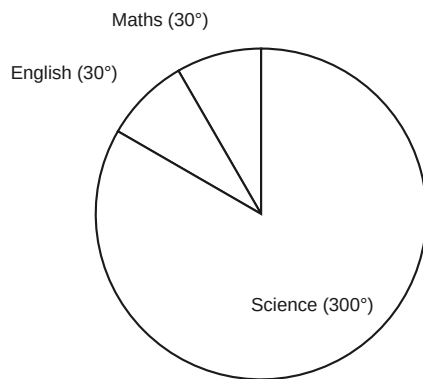
Q1. A survey asked 18 people about their favourite sport. Calculate the angle for each category, and draw a pie chart.

	Frequency	Angle
Football	6	
Tennis	2	
Swimming	10	

Statistics › Charts and Graphs › Interpreting Pie Charts (Foundation)

Topic 303 of 313 • id: pie_chart_interpret_F

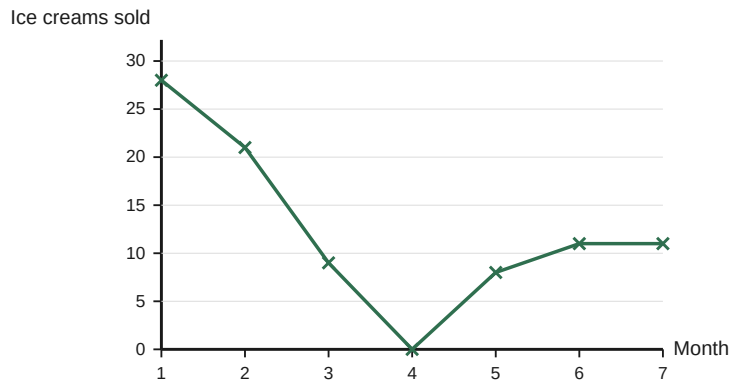
Q1. The pie chart shows the favourite subject of 12 people. What fraction of the total chose Science? Give your answer in its simplest form.



Statistics › Charts and Graphs › Time Series Graphs (Foundation)

Topic 304 of 313 • id: time_series_graph_F

Q1. The time series graph shows the ice creams sold over time. At which period was the value highest?



Statistics › Charts and Graphs › Constructing Scatter Graphs (Foundation)

Topic 305 of 313 • id: scatter_graph_construct_F

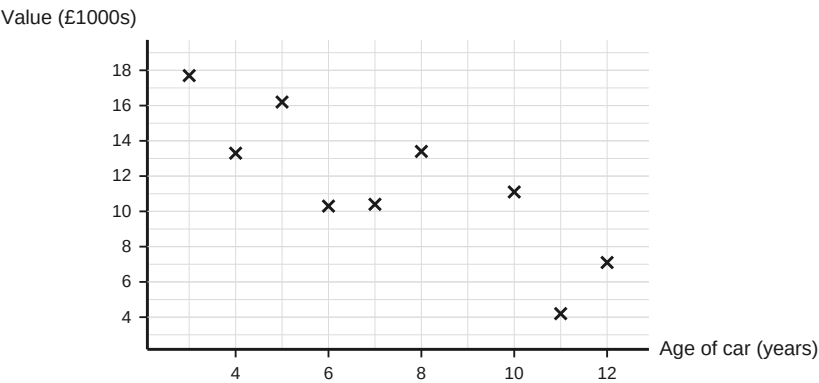
Q1. The table shows the midday temperature on a number of days and the number of ice creams a shop sold. Draw a scatter graph of this data, and describe the correlation.

	1	2	3	4	5	6	7	8	9
Temperature (°C)	12	13	15	16	17	19	20	25	29
Ice creams sold	46.6	50.5	46.0	57.5	65.6	75.0	62.4	81.3	103.5

Statistics › Charts and Graphs › Interpreting Scatter Graphs (Foundation)

Topic 306 of 313 • id: scatter_graph_interpret_F

Q1. The scatter graph shows the age of a number of second-hand cars and their value. Draw a line of best fit, and use it to estimate value (£1000s) when age of car (years) is 8.

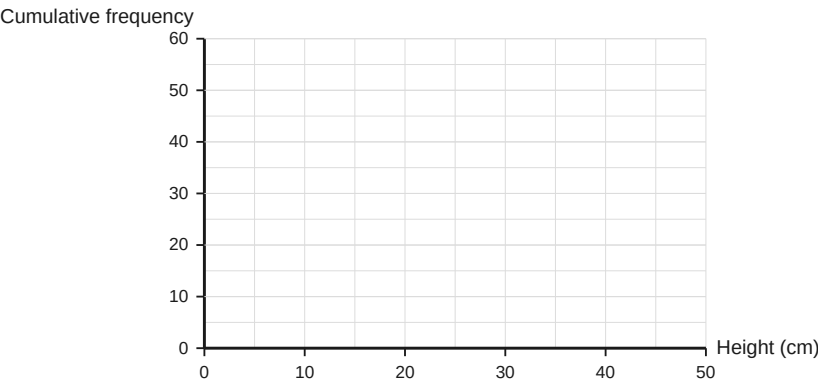


Statistics › Cumulative Frequency & Box Plots › Plotting Cumulative Frequency Graphs (Higher)

Topic 307 of 313 • id: cumulative_frequency_plot_H

Q1. The table shows the heights (in cm) of a group of plants. Draw a cumulative frequency graph to show this information.

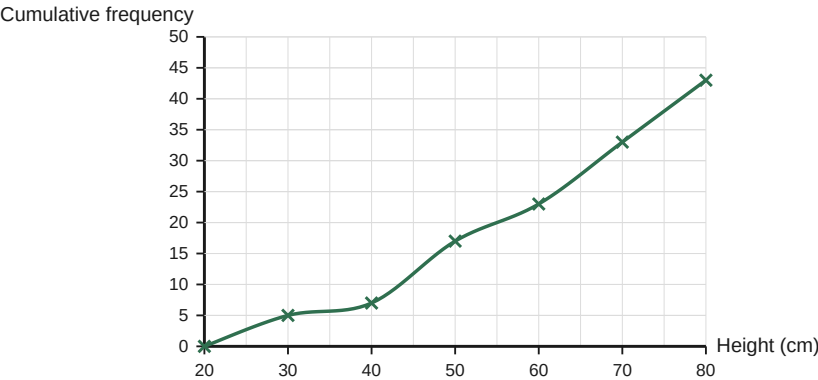
	Frequency
0-10	15
10-20	10
20-30	12
30-40	2
40-50	15



Statistics › Cumulative Frequency & Box Plots › Interpreting Cumulative Frequency Graphs (Higher)

Topic 308 of 313 • id: cumulative_frequency_interpret_H

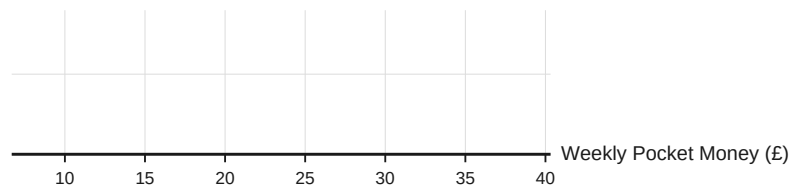
Q1. The cumulative frequency graph shows the heights (in cm) of a group of plants. Use the graph to estimate the interquartile range.



Statistics › Cumulative Frequency & Box Plots › Constructing Box Plots (Higher)

Topic 309 of 313 • id: box_plot_construct_H

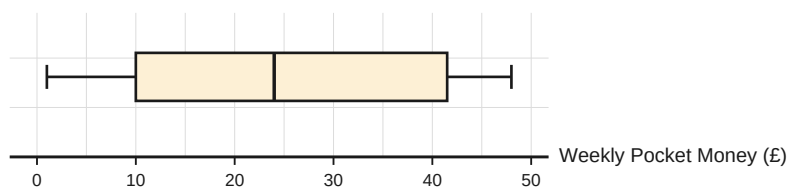
Q1. Here are seven weekly pocket money (£): [9, 15, 19, 24, 26, 30, 38]. Draw a box plot for this data.



Statistics › Cumulative Frequency & Box Plots › Interpreting Box Plots (Higher)

Topic 310 of 313 • id: box_plot_interpret_H

Q1. The box plot shows weekly pocket money (£). Find the interquartile range.

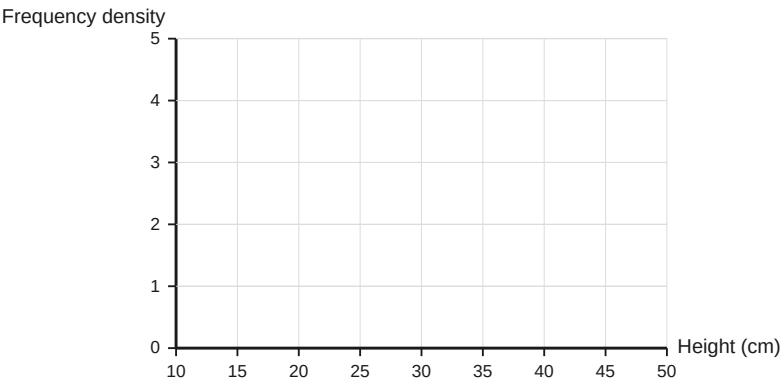


Statistics › Histograms › Plotting Histograms (Higher)

Topic 311 of 313 • id: histogram_plot_H

Q1. The table shows the heights (in cm) of a group of plants. Complete a frequency density column and draw a histogram.

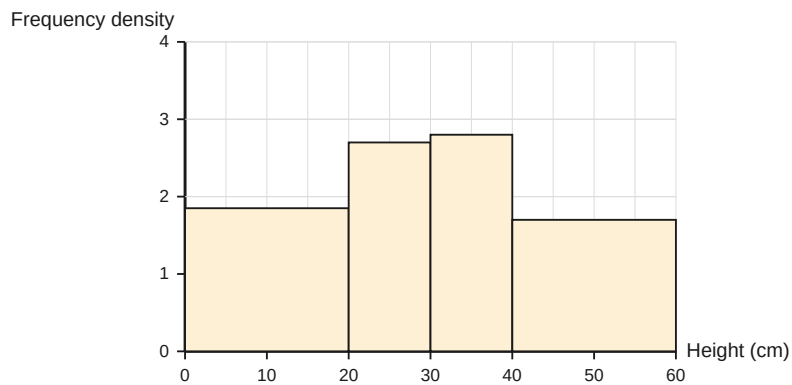
	Frequency
10-15	6
15-35	18
35-45	35
45-50	18



Statistics › Histograms › Interpreting Histograms (Higher)

Topic 312 of 313 • id: histogram_interpret_H

Q1. The histogram shows the heights (in cm) of a group of plants. Which class has the highest frequency?



Statistics › Sampling and Populations › Sampling and Populations (Foundation)

Topic 313 of 313 • id: sampling_methods_F

Q1. A school has 393 students: Year 9: 220, Year 7: 173. A stratified sample of 55 students is taken. How many students from Year 7 should be in the sample?